Panithi Makthiengtrong

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Linkedin: /in/betich
GitHub: github.com/betich

About Me

When I was 15, I got in to programming for the first time.

I took a Java course hoping to mod *Minecraft*. Since then, I've been hooked on the process of *creating* things and sharing things, which I explored in various different type of forms.

I recently complete a bachelor's degree in Robotics and Artificial Intelligence engineering at Chulalongkorn University (*Class of 2026*), and now I work across the full process: software engineering, embedded systems, graphic design, marketing, and whatever else a project demands. I'm drawn to new frontiers and the spirit of building something from nothing, then refining it until it's exactly what it should be.

Here, I hope this collection of memories offers a sufficient portrait of who I am, what I've built, and what I stand for.

github.com/betich

betich.me

resume.betich.me

My Skills

- Software Engineering
- Embedded Systems
- Robotics
- System Design
- Game Development
- Audio/Visual Programming
- DevOps
- Artificial Neural Networks
- Video Editing
- Social Media / Digital Marketing
- Mechanical Design
- Project Management
- Data Analysis
- Graphic Design

Programming Languages: JavaScript, Node.js, TypeScript, Python, Golang, C++, C (Arduino)

Libraries & Frameworks: ReactJS, NextJS, TailwindCSS, ExpressJS, Svelte, Pytorch, Tensorflow, Astro, Ionic React, Expo

Robotics Tools: ROS2, MuJoCo, Arduino, Raspberry Pi, Raspberry Pi Pico, I2C/SMBus, BLE, RF, DMX, PCB Design (KiCad, EasyEDA), Fusion 360, Motor control

Project Showcase

01 Robotics and Embedded Hardware

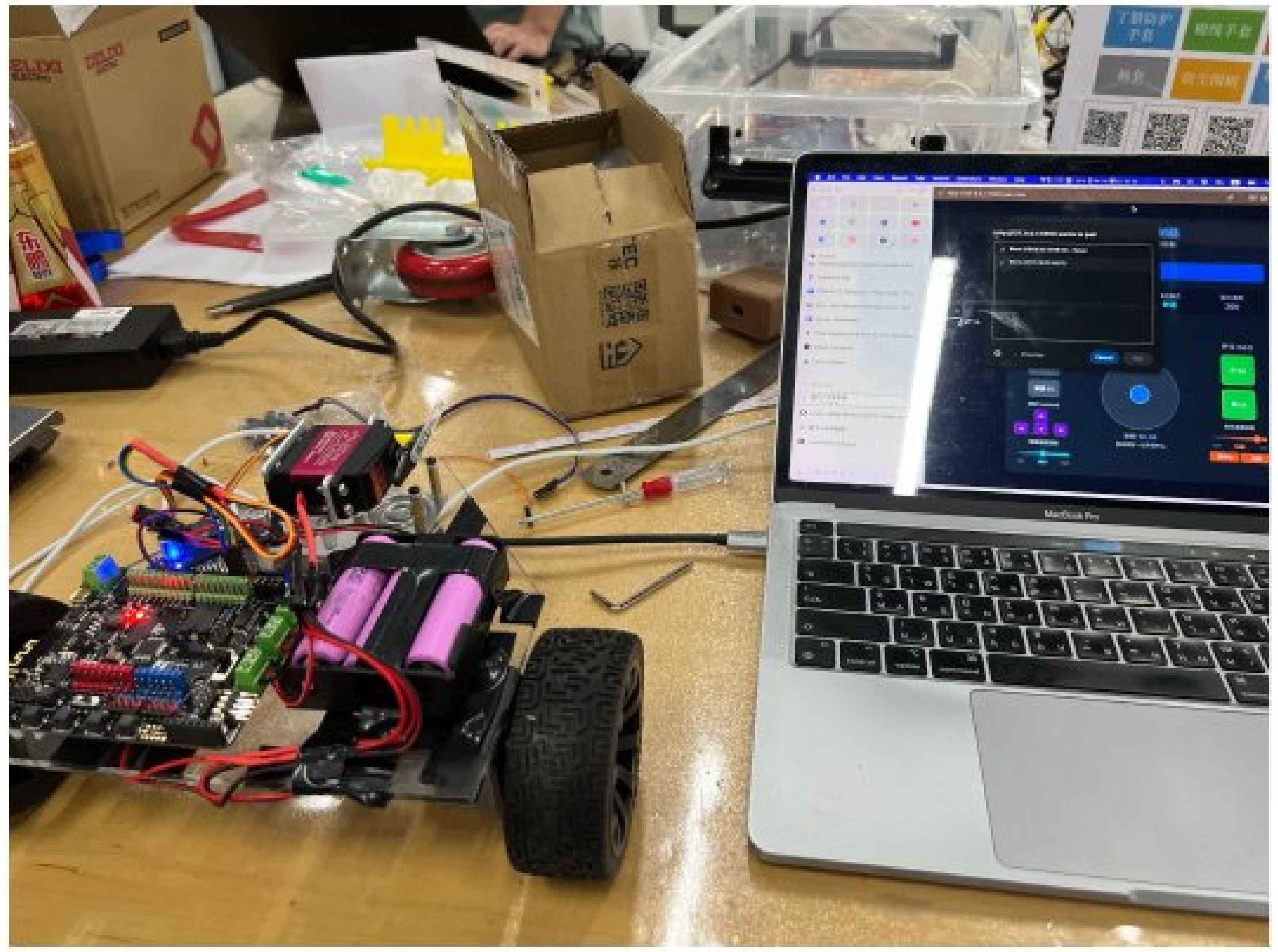
02 Software Engineering

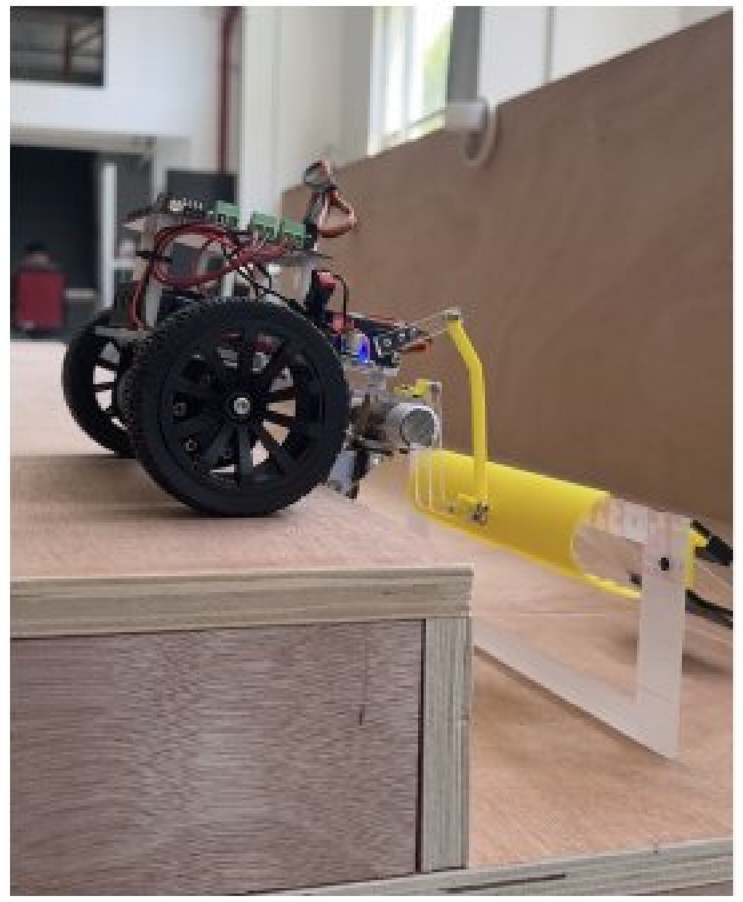
03 Graphic Design and Marketing

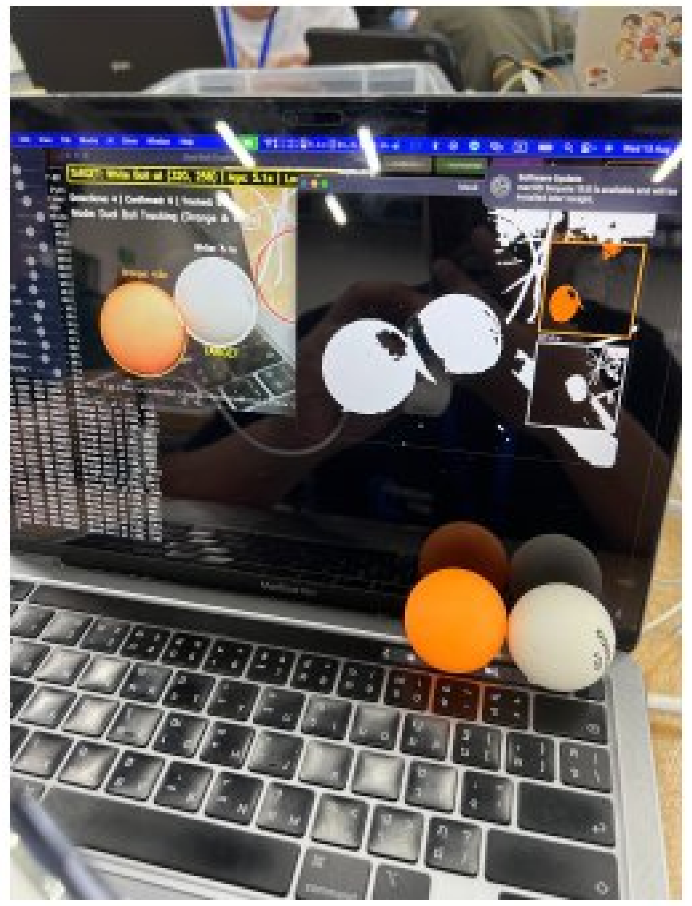
01 **Robotics and Embedded Hardware**

02 Software Engineering

03 Graphic Design and Marketing





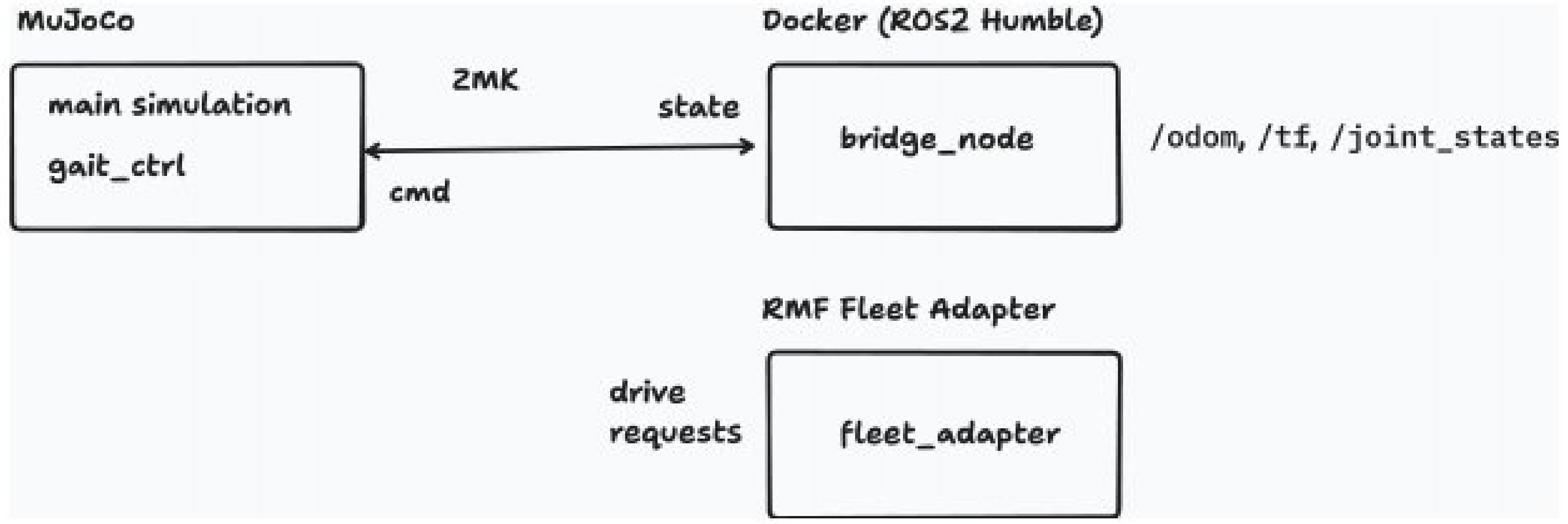


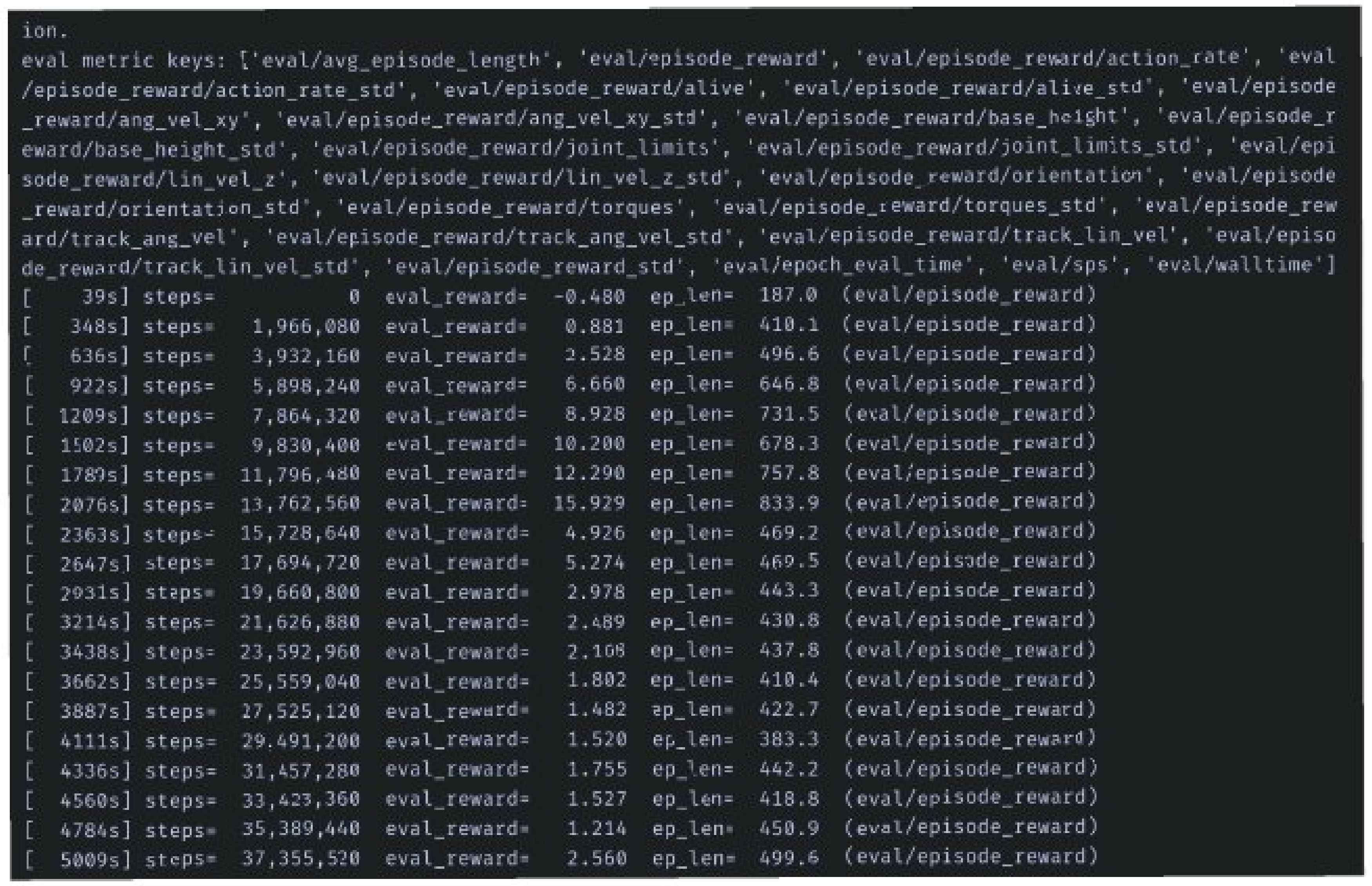
Skills used in this project

- Ros2 and MuJoCo for simulation
- Arduino (C++) for controlling motors and communicating with web interface and an RF receiver
- Web control interface controlled by BLE

IDC RoboCon 2025

My team won ***first place*** in IDC RoboCon, an international robot design competition hosted in Shanghai, China with participants from Thailand, China, Japan, South Korea, Egypt, and India. We were tasked to build a builder and a “mission scorer” robot in the timespan of about two weeks - completing the design process, fabrication, and programming along the way.





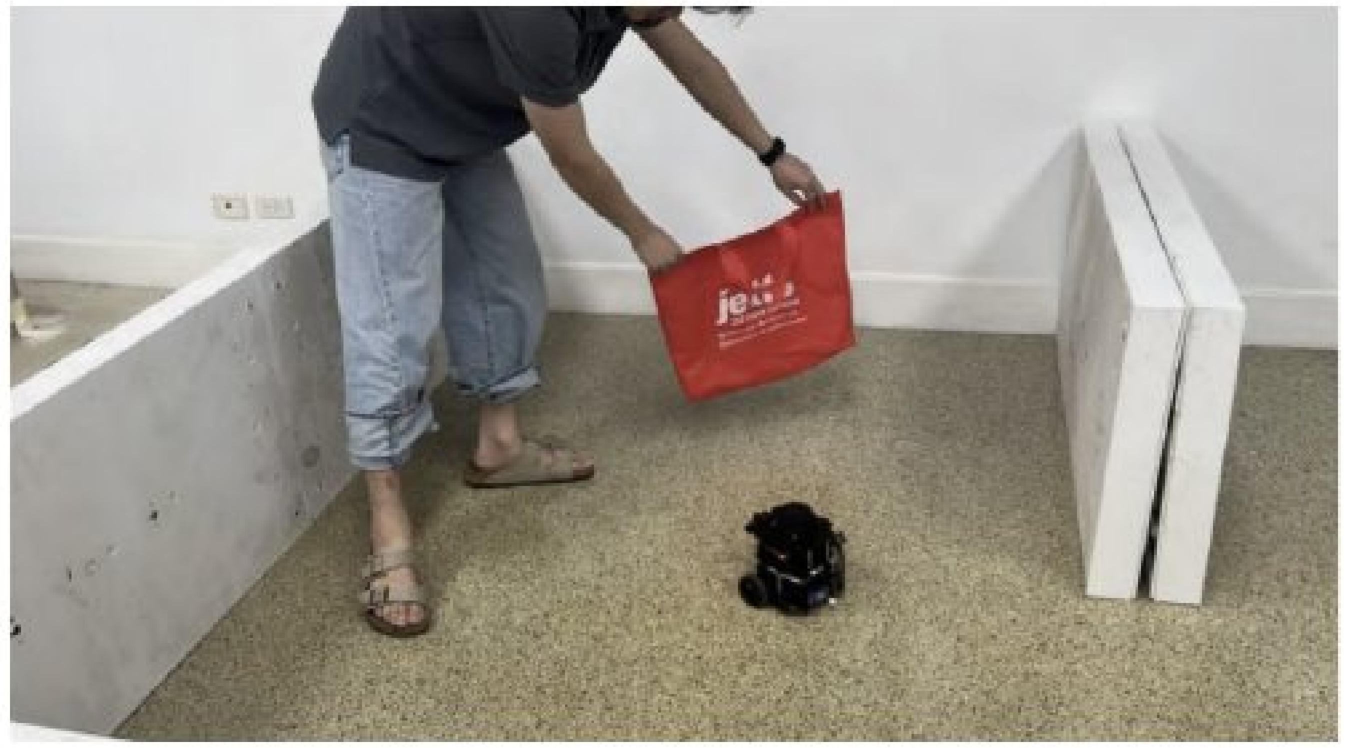
Skills used in this project

- MuJoCo for simulation
- JAX via MuJoCo XLA (MJX) for Deep Learning
- ROS2 and RMF for hardware connection

GitHub Link

Dog Robot Simulation Control

This project uses a robotic dog simulation to test kinematics, machine learning for movement control, and hardware integration via ROS2, MuJoCo, and RMF. Using the ANYbotics ANYmal as a starter model, I experimented with gait control by adjusting reward and penalty functions to make the robot walk, run, turn, and strafe. This work acts as an exploratory first step for a custom robotics project I am developing with friends.



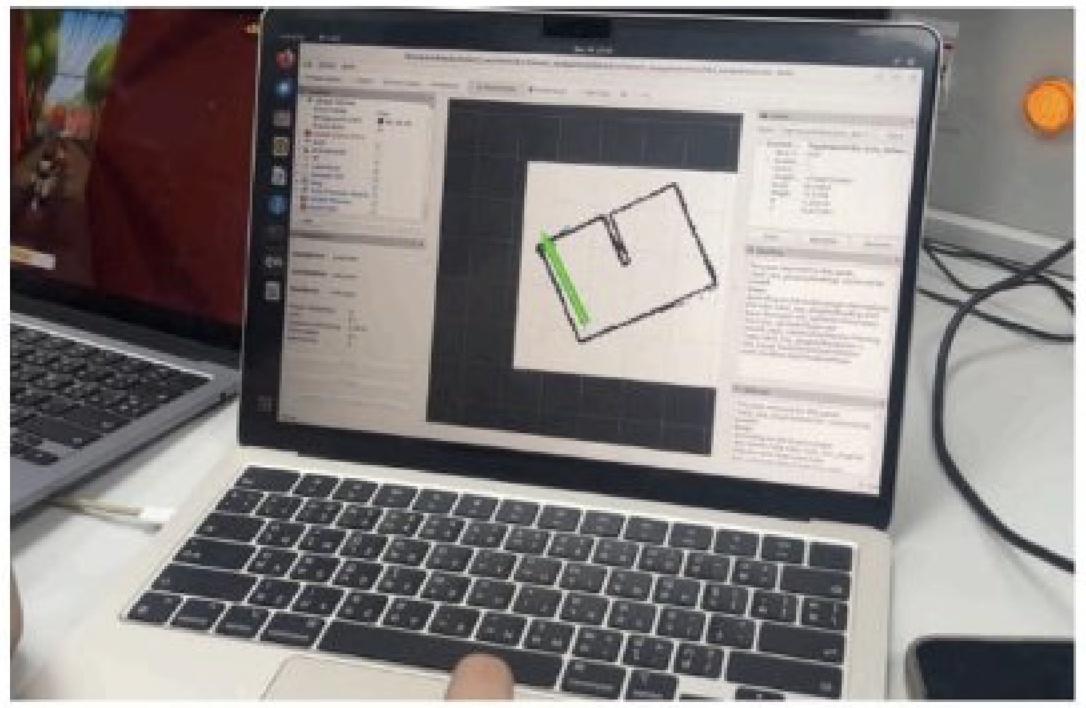


Figure 4 : Path estimation using RViz.

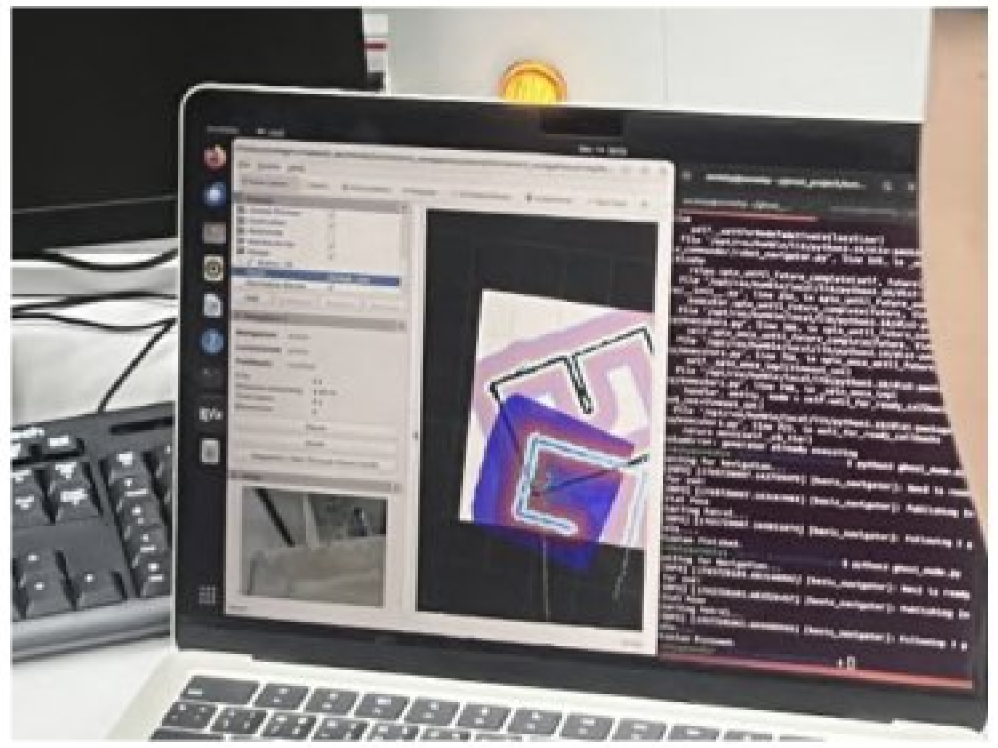
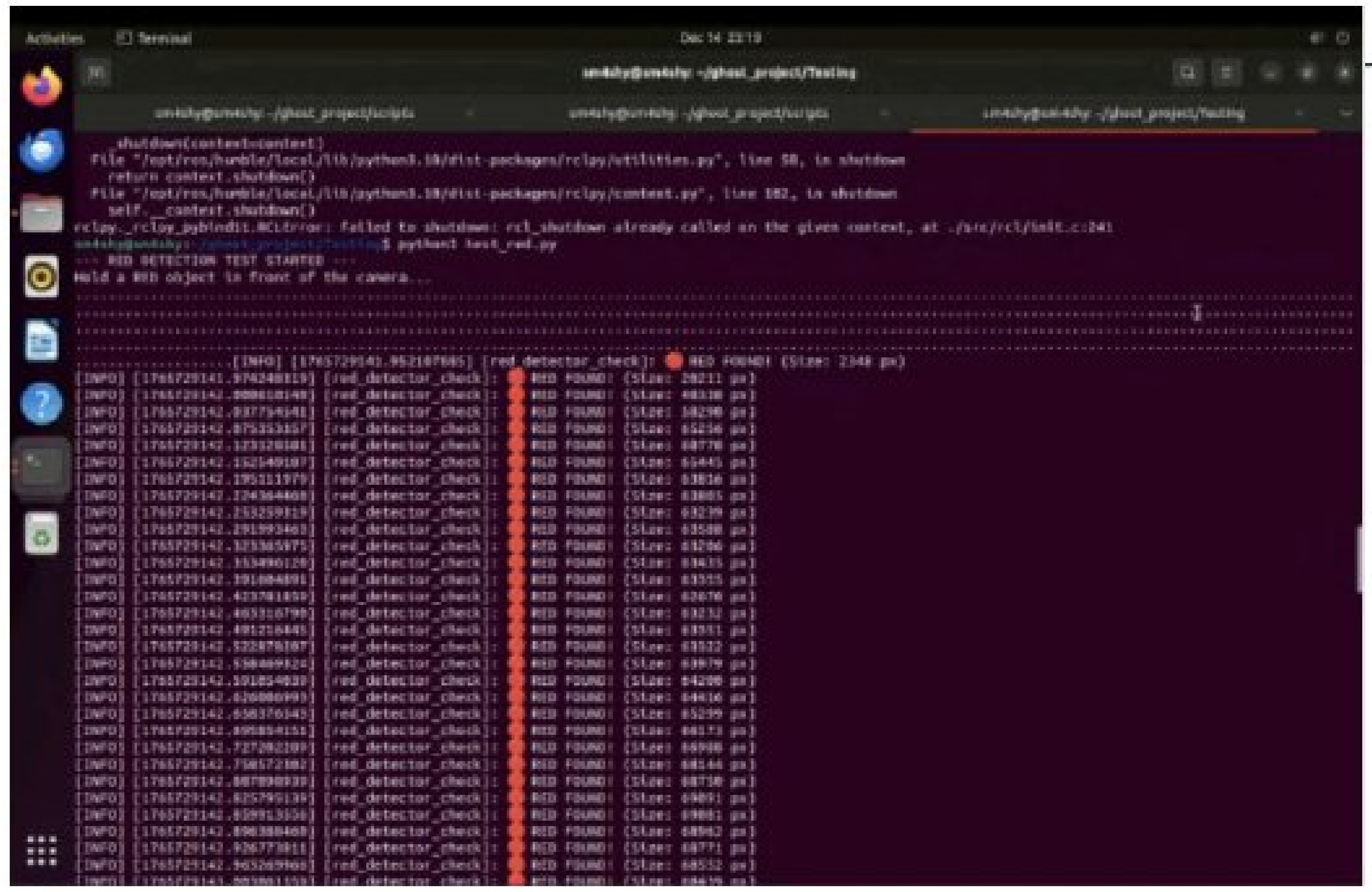


Figure 5 : Real-time visualization of robot mapping and path planning using RViz

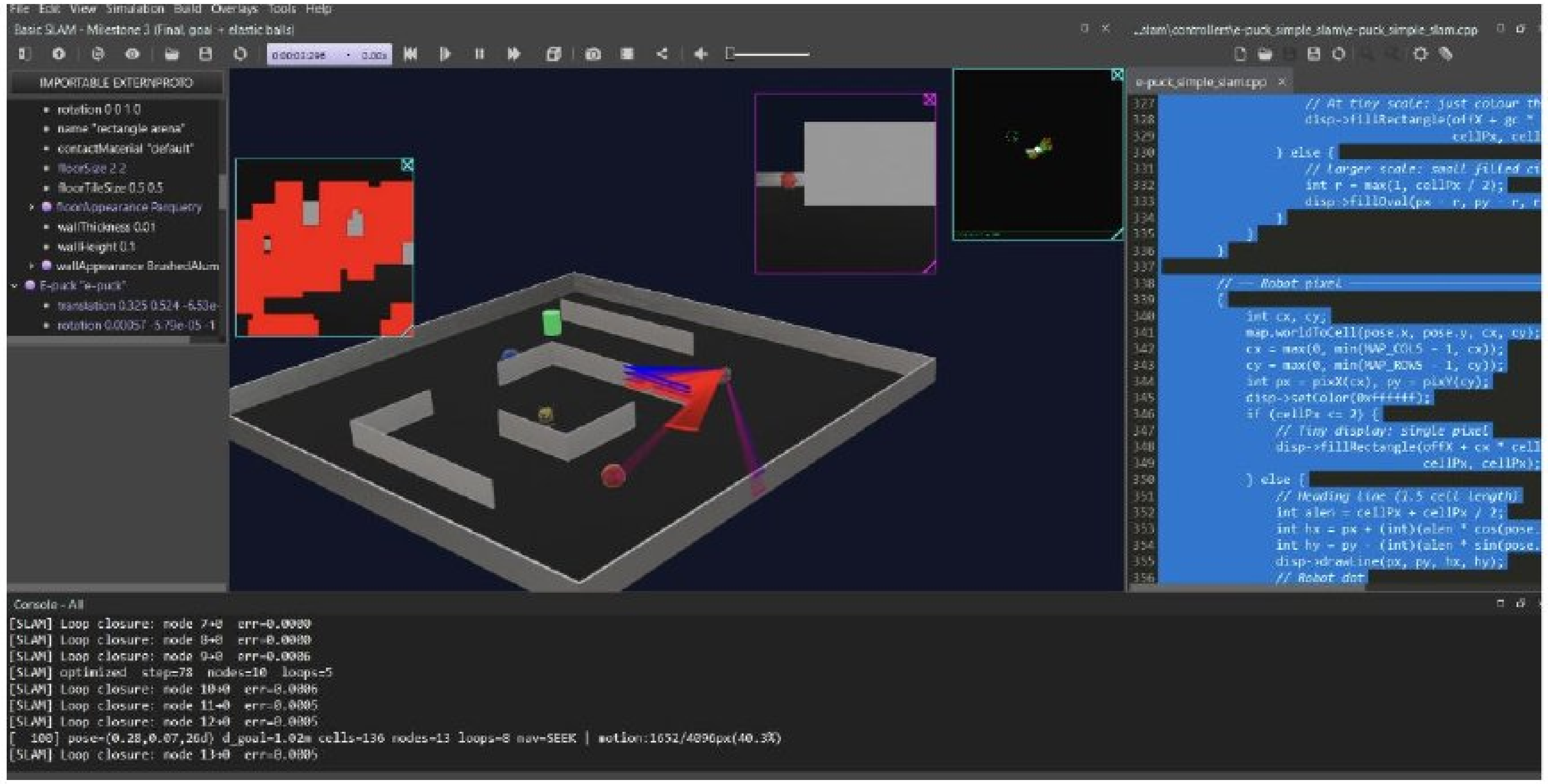


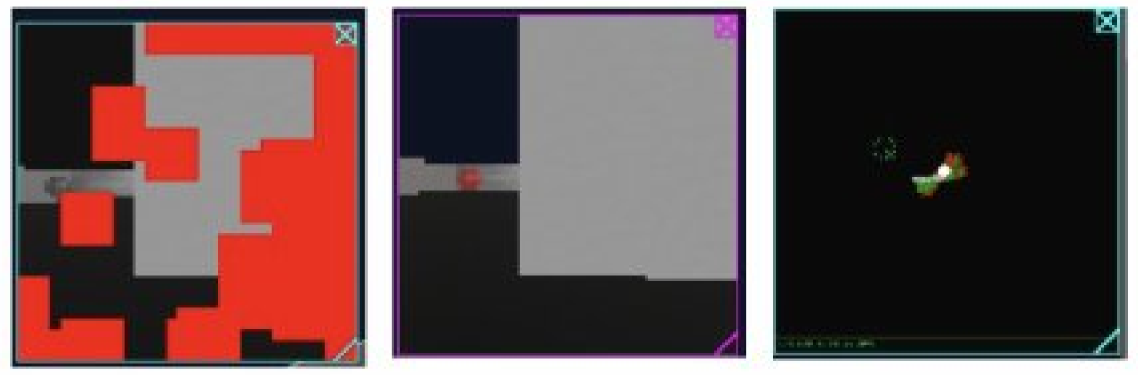
Skills used in this project

- ROS2 for hardware control and navigation
- Mapping and path planning using RViz
- Image processing (Finding red objects)

Advanced Mobile Robots Class Project

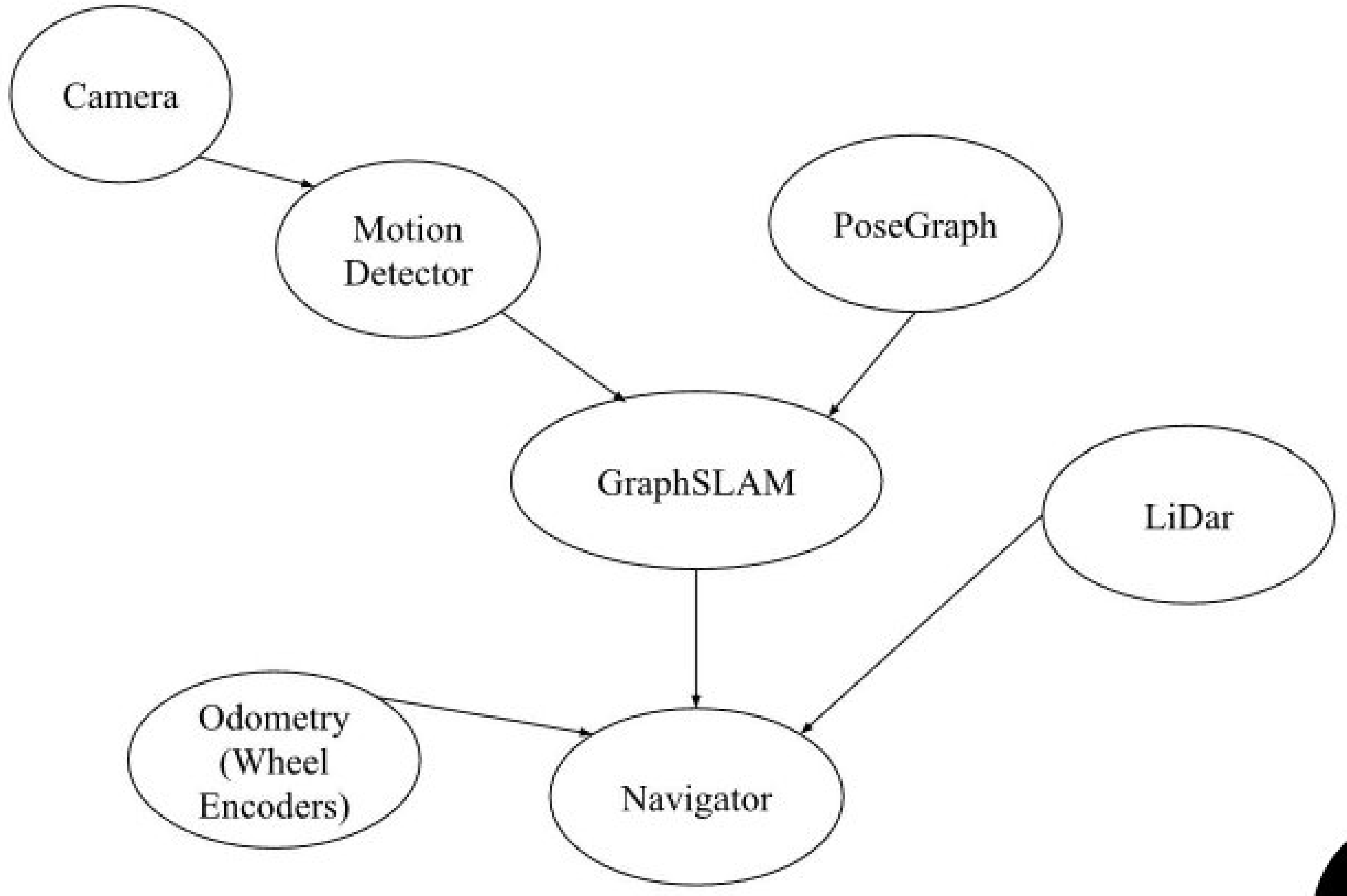
My group built a robot that navigates its surroundings to patrol and seek out a target, by making a turtlebot map out its surroundings using SLAM via Lidar and using RViz for path planning and mapping, and creating a target spot for the turtlebot to navigate to, then use image processing to detect any threats that it sees, which in this case, is a red object.





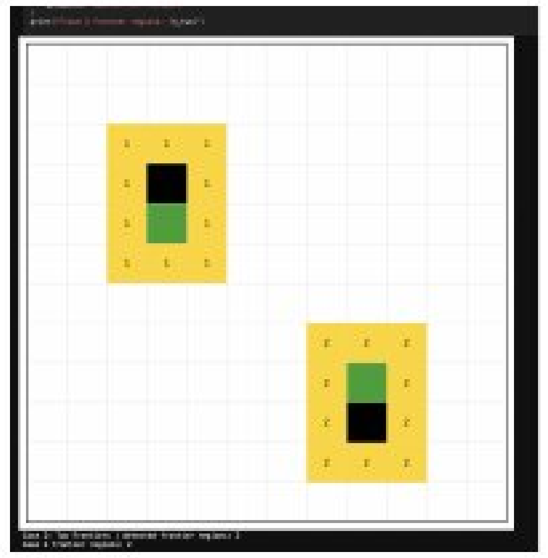
Map Display Layer: Green = Free space Brown/Red = Occupied space Gray Dots/Lines = Historical pose nodes and odometry edges Cyan Lines = Loop closure connections White Dot + Line = Current robot position and heading

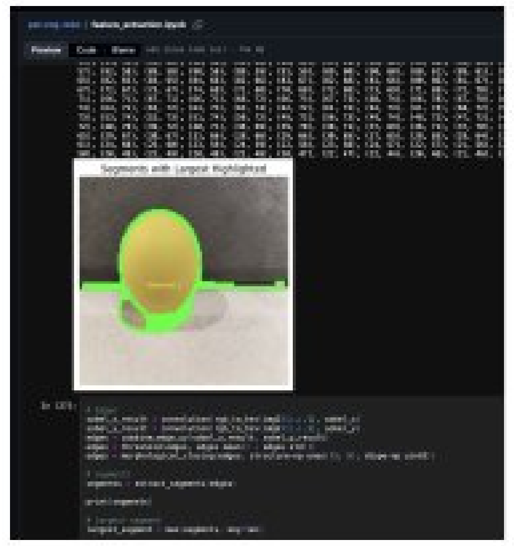
Motion Vision Display: Red Pixels = Dynamic/moving objects detected via optical flow Grayscale = Static environment

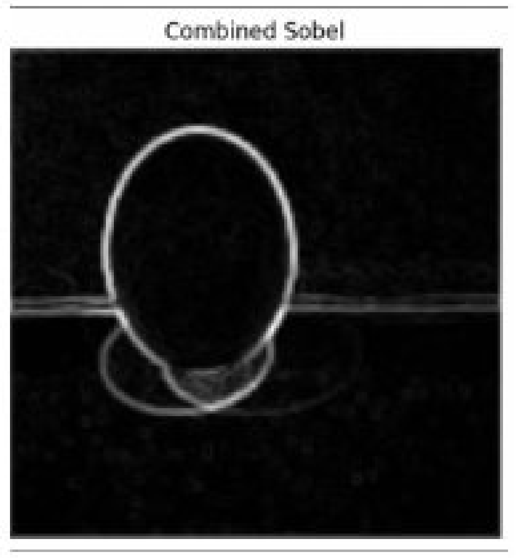


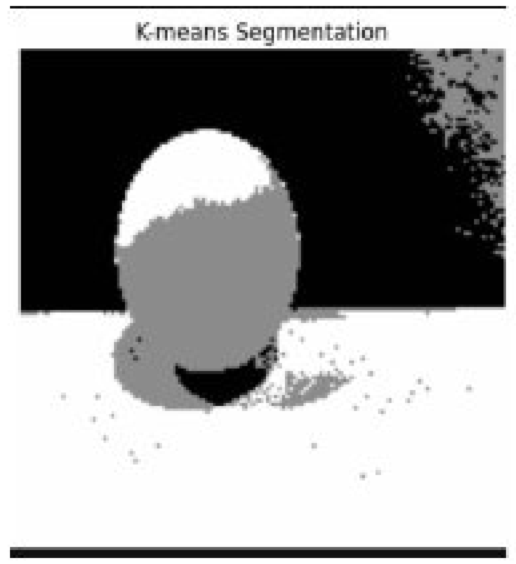
Skills used in this project

- WeBots for Robot Simulation
- Scripting using C++, Python, Lisp
- Image processing algorithms
- Optical flow and Navigation





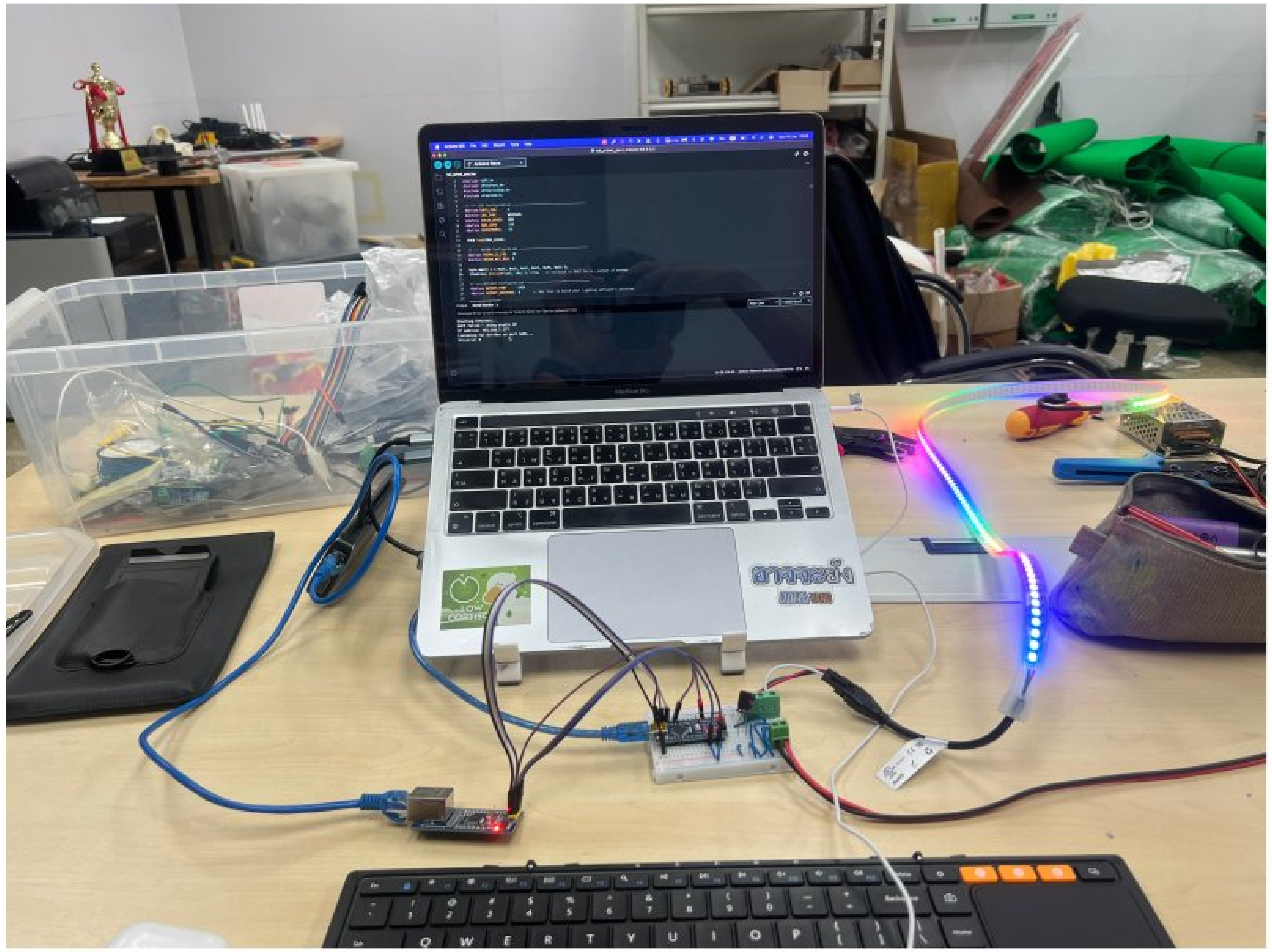


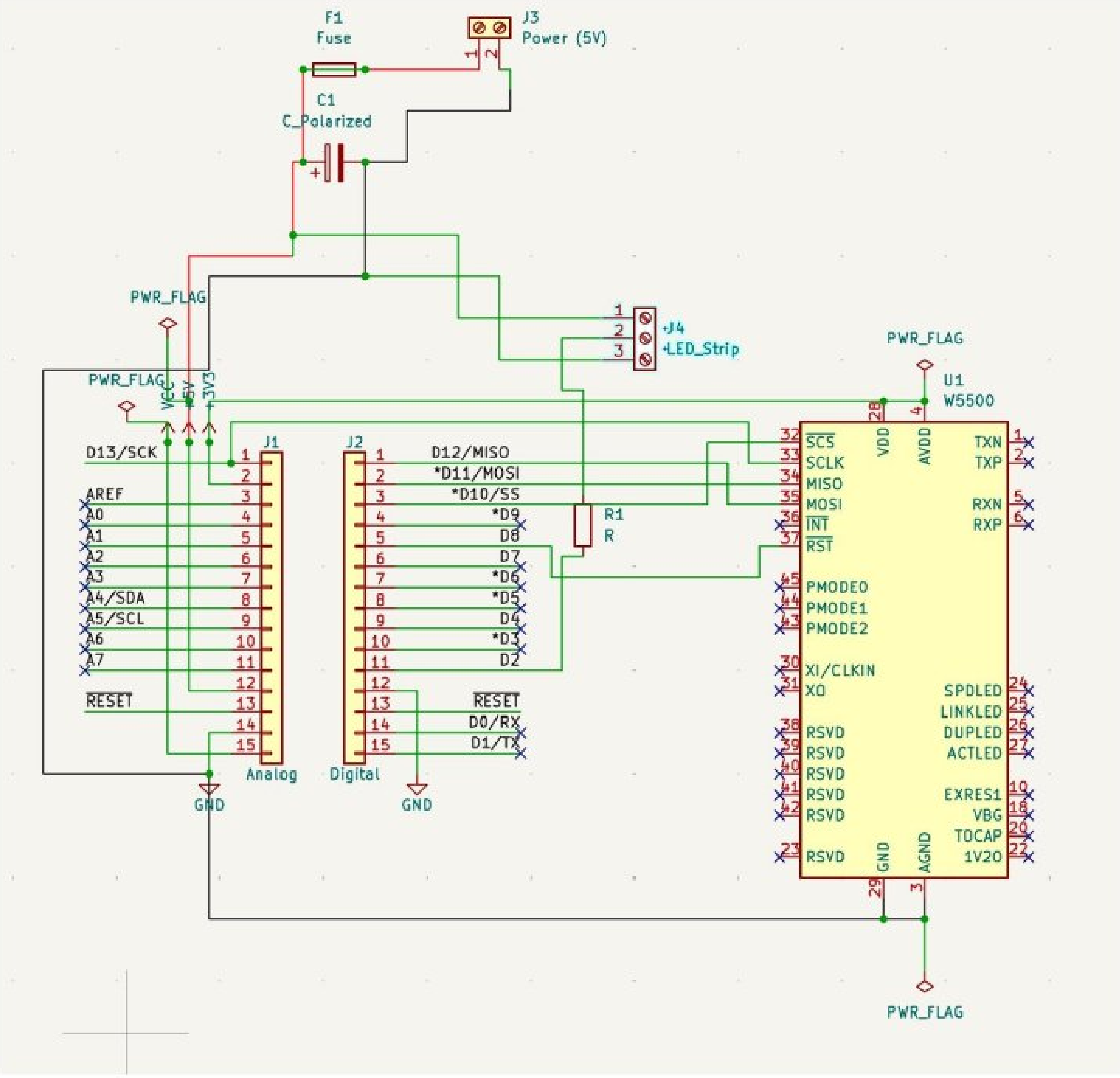


GitHub Link
(Code Snippets)

Perceptive Cognitive Robot Class Project

This project is done throughout the Perceptive Cognitive Robots class, where it explores image processing, SLAM and optical flow for robots.The vision and SLAM system is built using C++, interfacing directly with Webots device APIs (Camera, Lidar, Motor, PositionSensor, Display) to process multi-modal sensor data and render live, multi-layered diagnostics to the Webots simulation screen. The system is built on Graph-based SLAM, Frontier-Seek Navigation, Occupancy, and Object Detection.

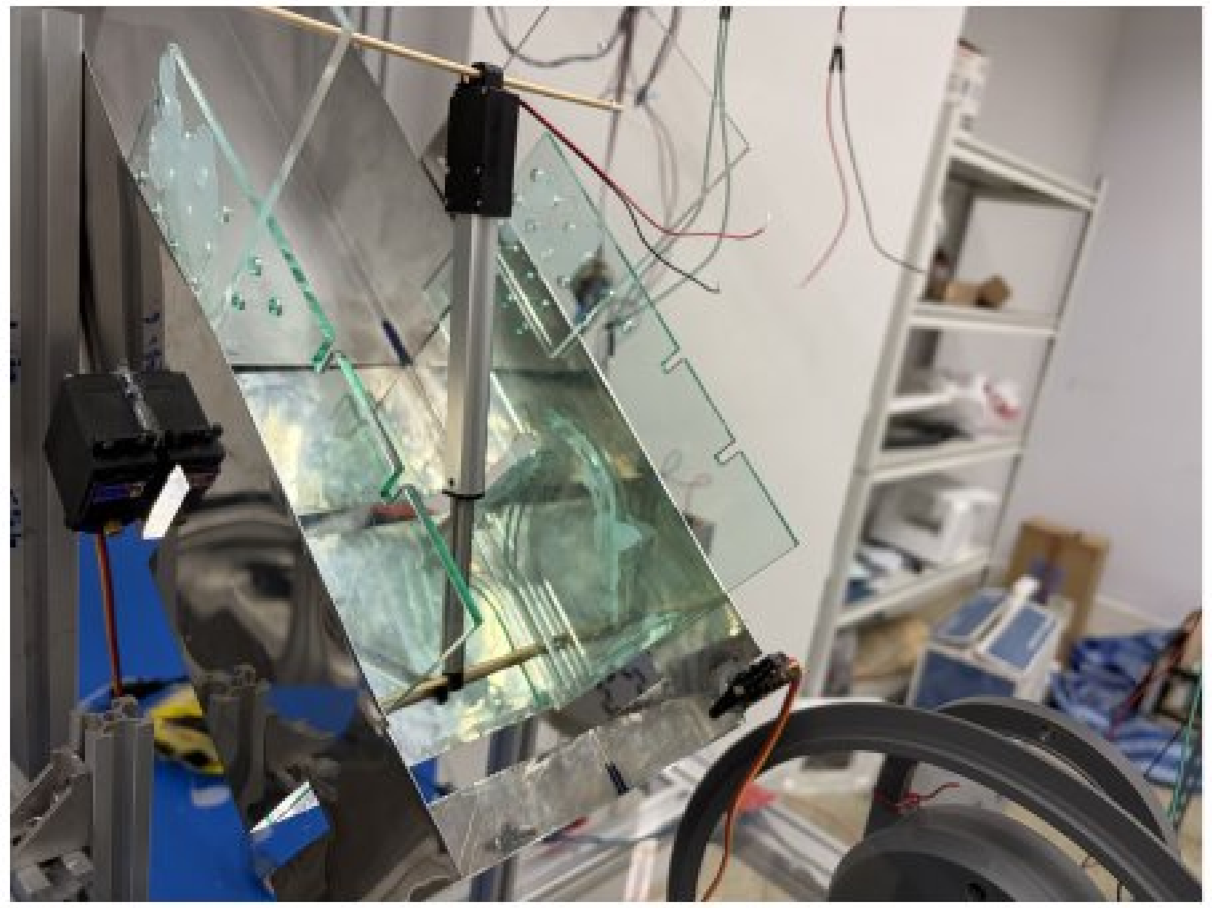


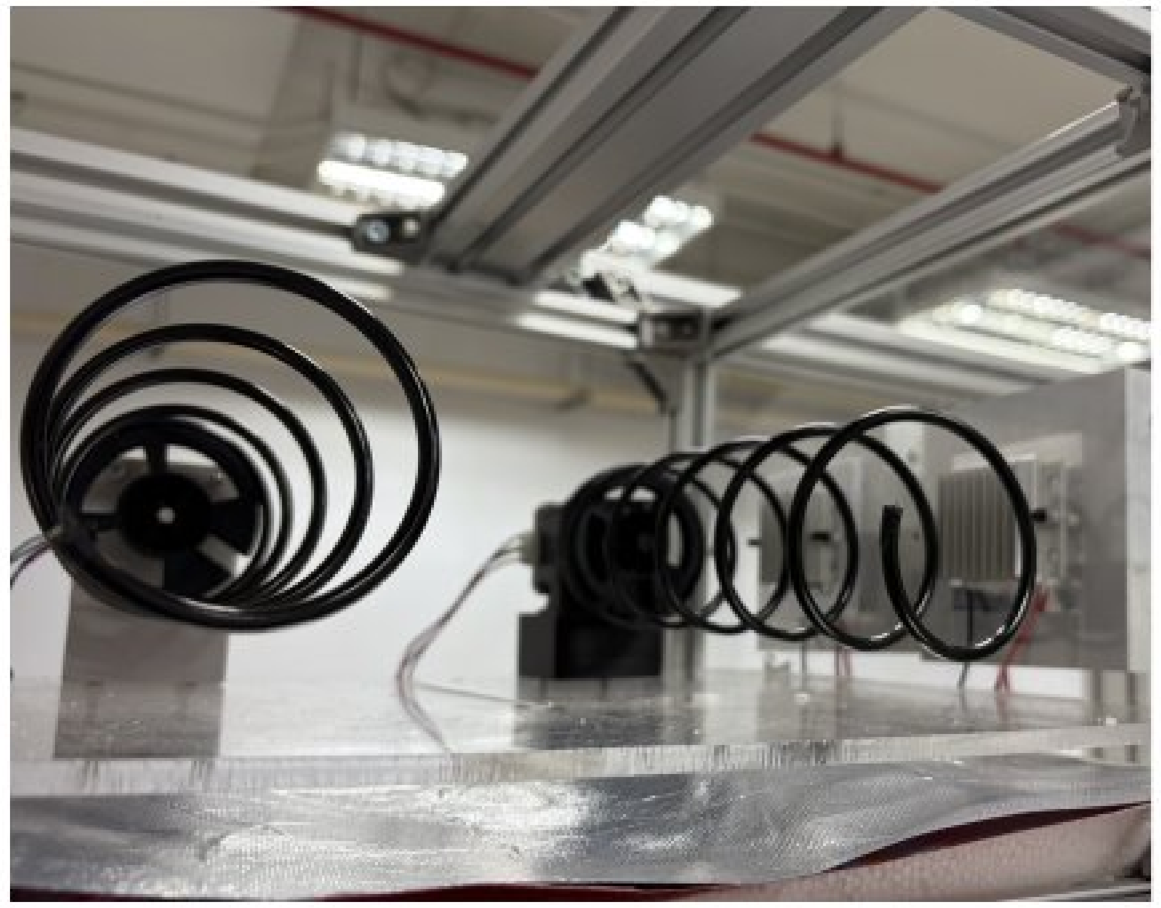


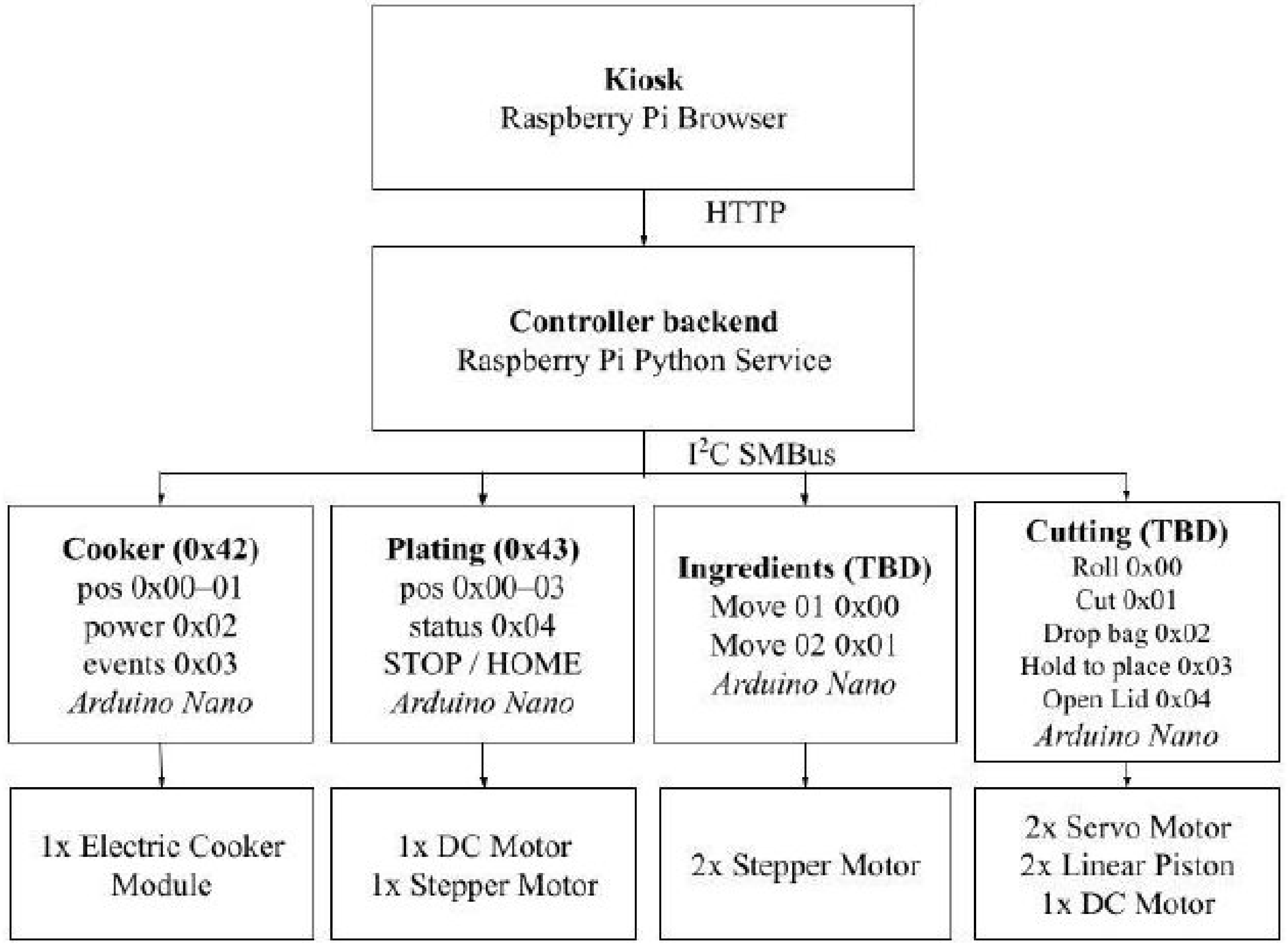
LED Strip ArtNet Controller

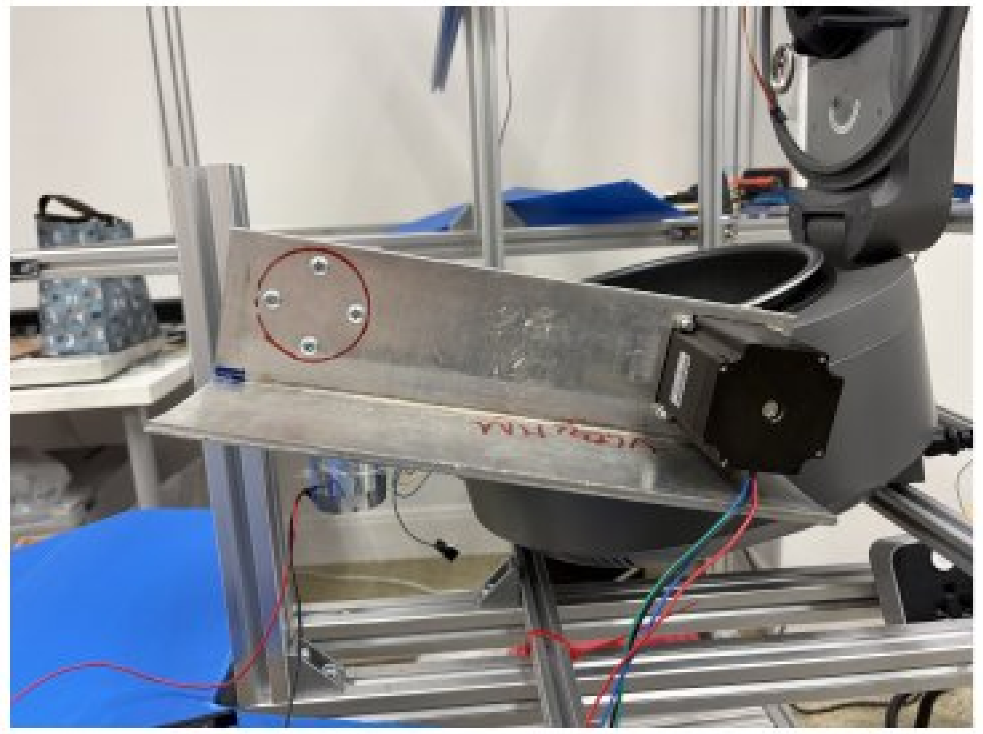
Role: Software, Electronics

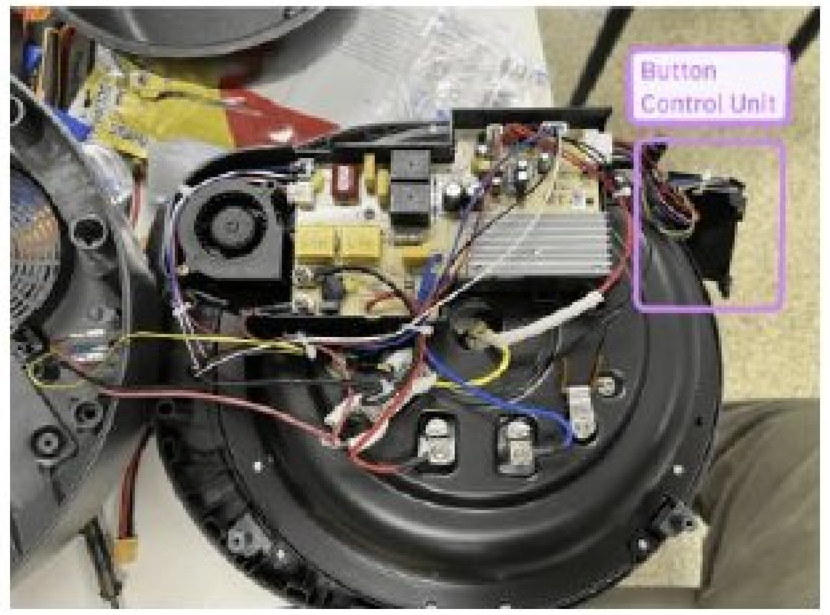
This is a small side project I created to control LED strips using Ethernet via a protocol via ArtNet, which artists can use software like TouchDesigner to communicate for it to display complex rendered graphics. This project stemmed from a curiosity from me and a visual artist friend who is seeking a cheaper way to control LED strips using DIY components while still supporting widely used protocols. Plans for expanding its capability and modularly adding more components such as DMX (XLR pins) are in the board.











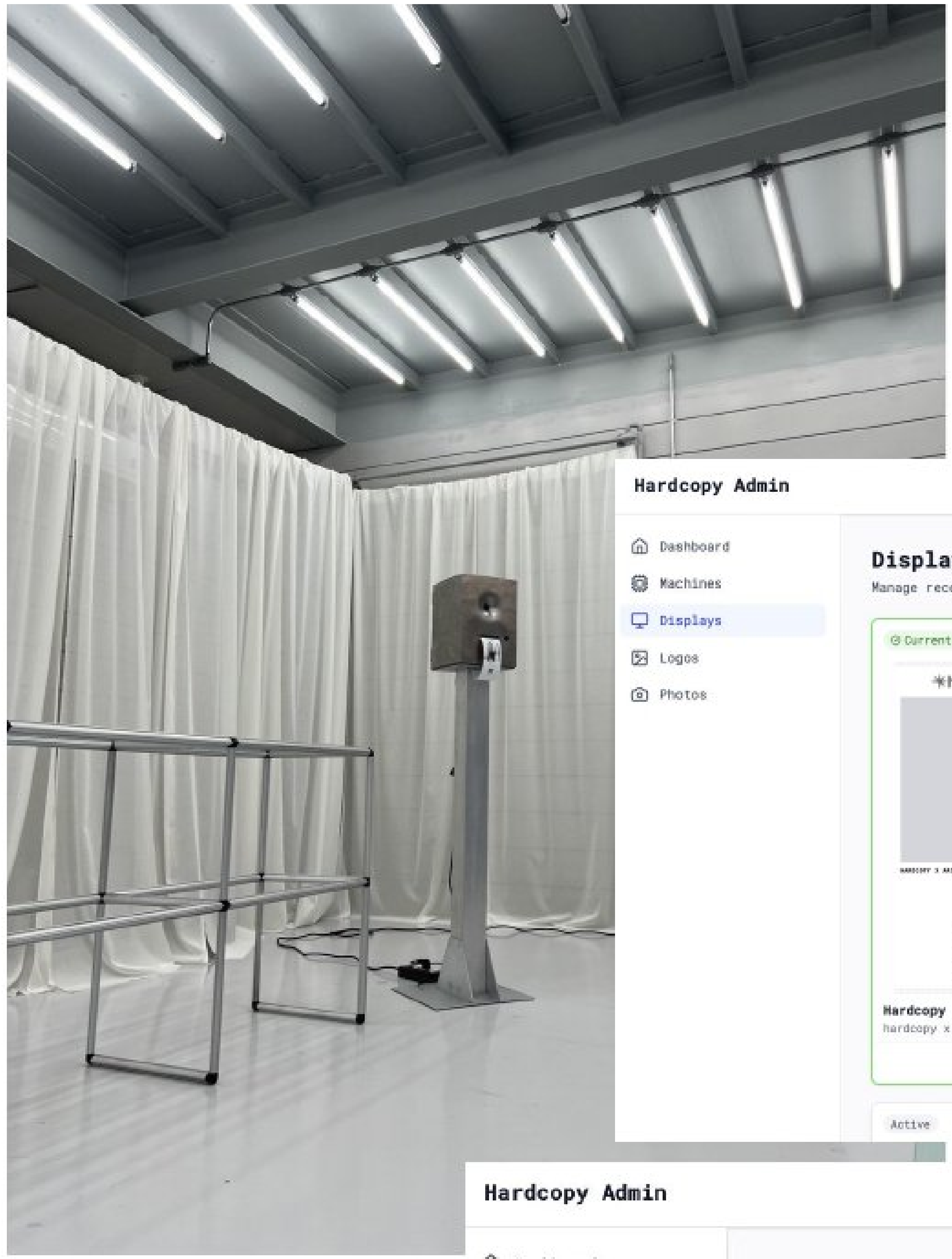
Hacked →

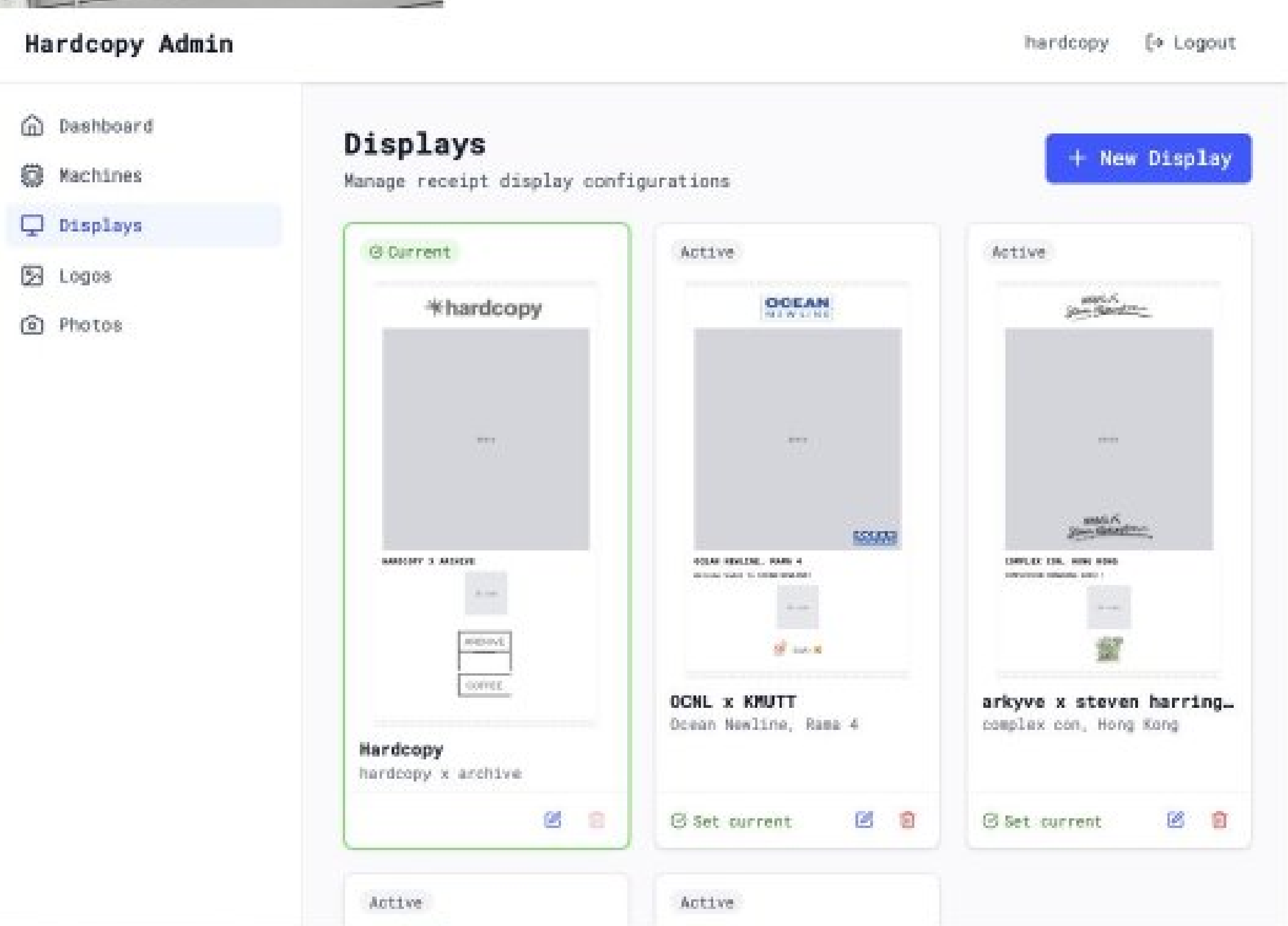
Tastebox

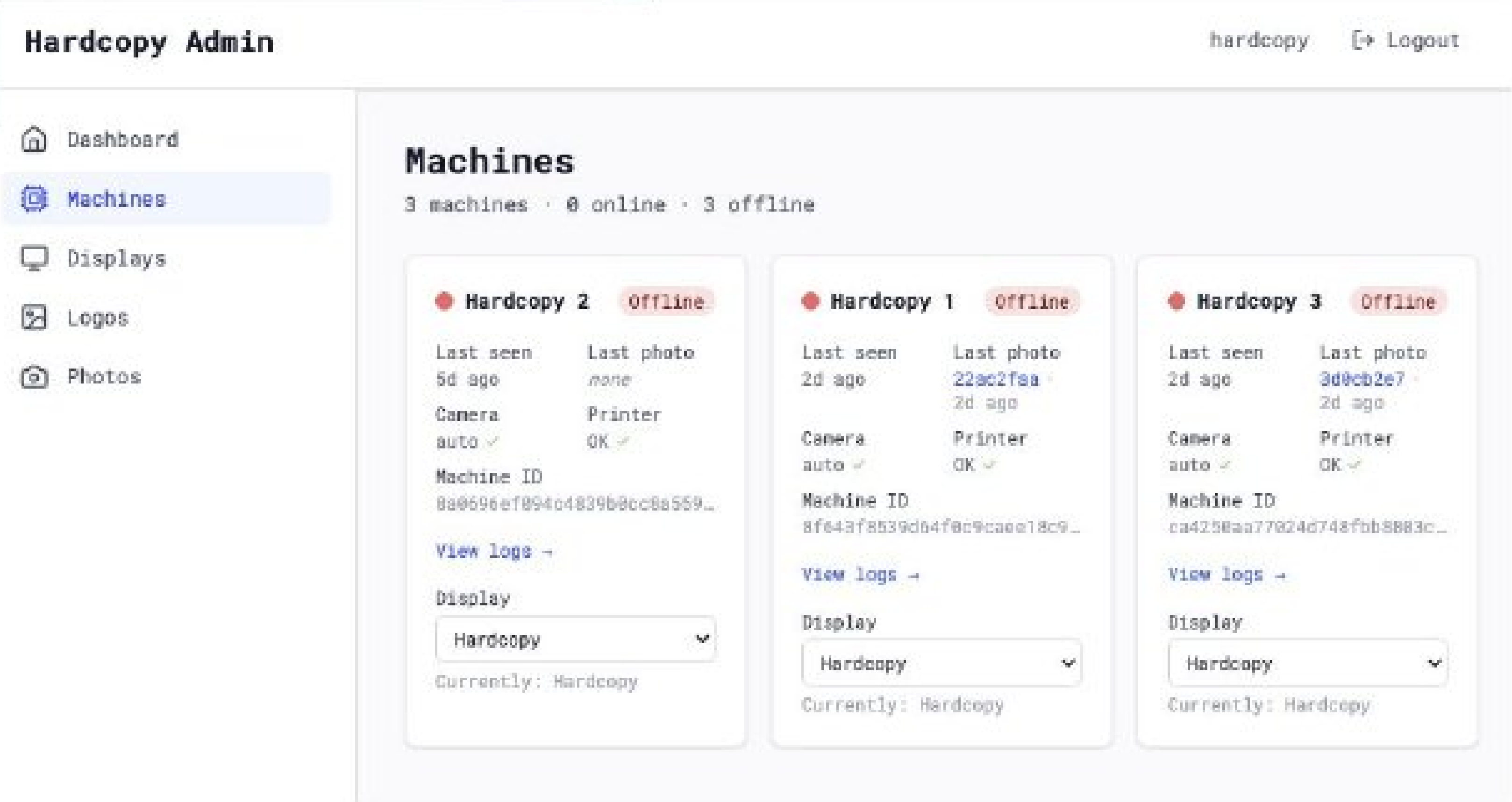
Role: Software, Mechatronics Engineer

GitHub Link **YouTube Link**

My final year engineering project is Tastebox, an automated robotic cooker. It featured four submodules each powered by a microcontroller, including the ingredients dispenser, bag cutting module, cooker module (which is programmed by replacing the button controlling module with a microcontroller — emulating the signals generated by the rotary encoder and the buzzer), pot controller and lid opener (robotic arm). All submodules are controlled an I2C bus with a Raspberry Pi.





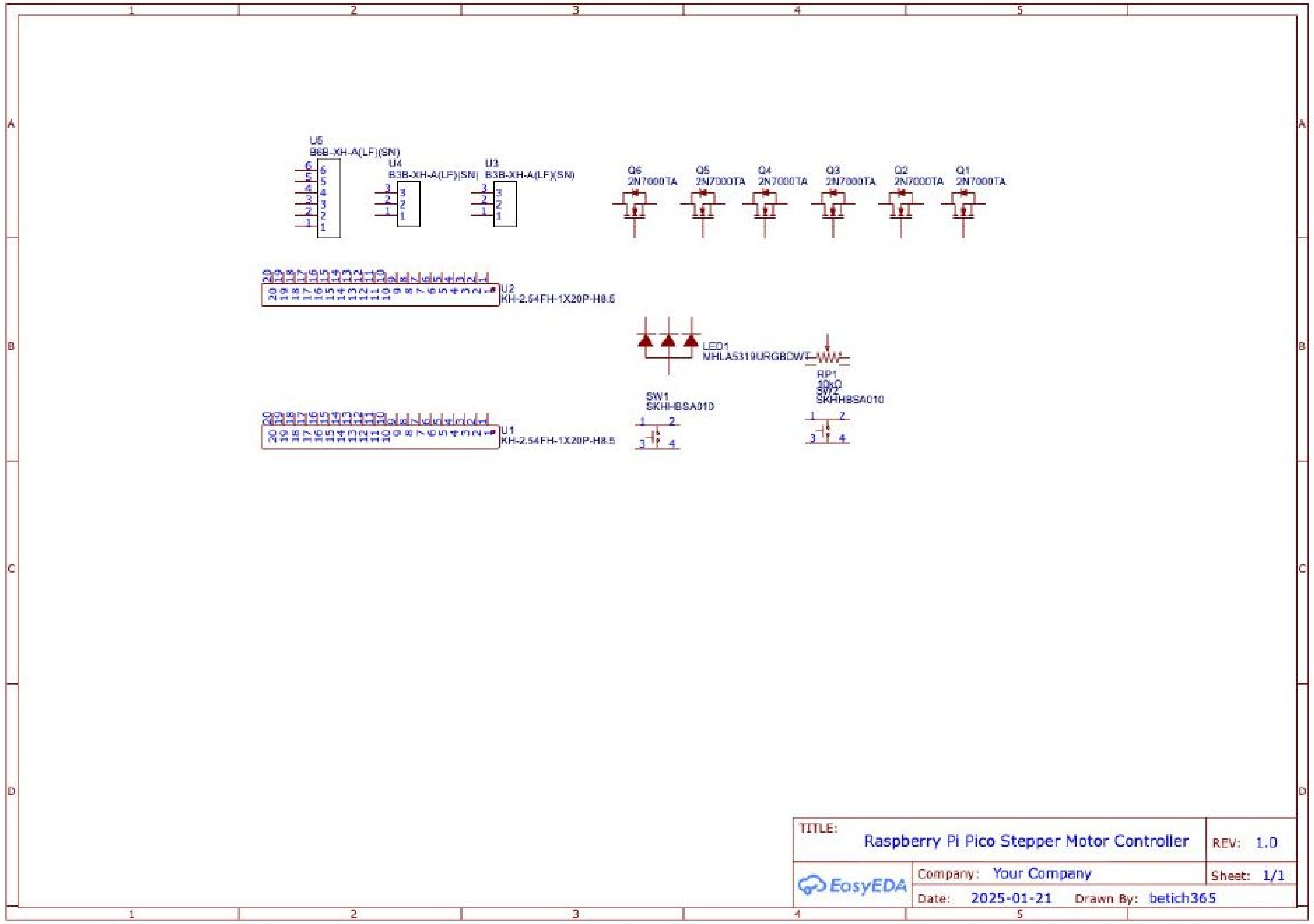


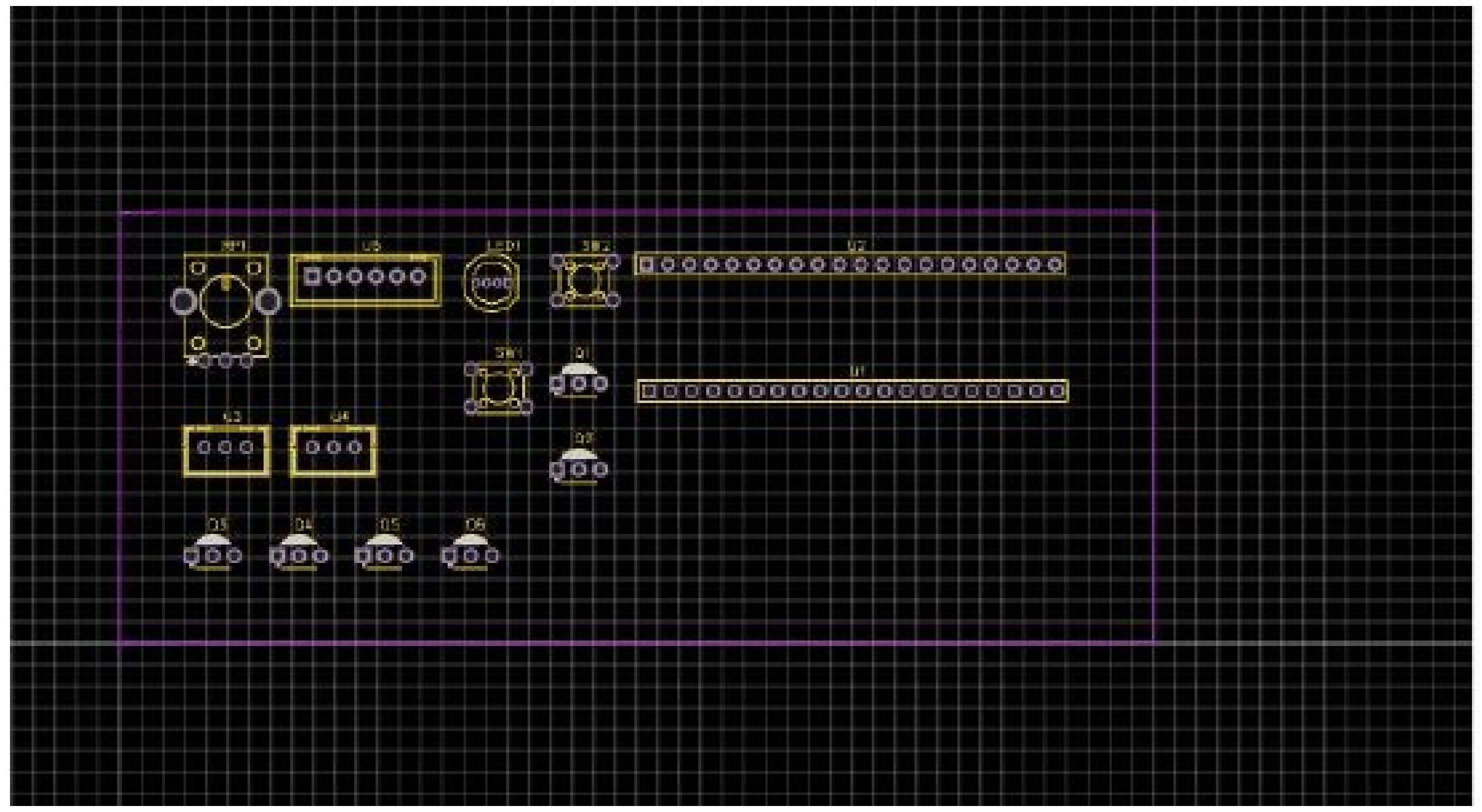
Hardcopy

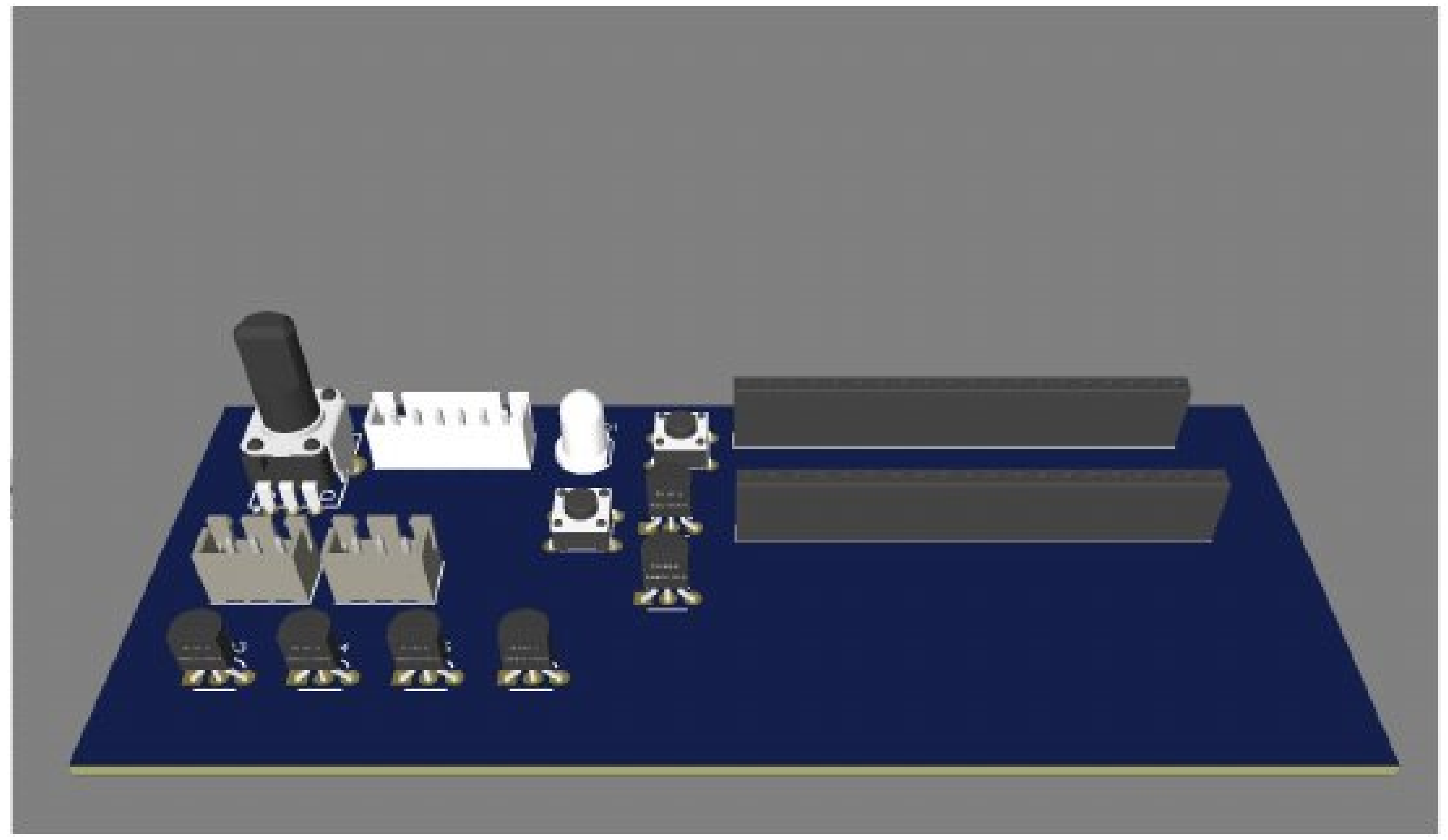
Role: Engineer

Hardcopy is a thermal printer photo booth project which operated by a Raspberry Pi connected to a camera and a printer. I developed the underlying code of the embedded hardware managing several machines, along with developing a dashboard to manage and monitor the machines and layout of the printed design.

The machines are installed in cafés and events with versioning and remote control on all machines to operate everything .

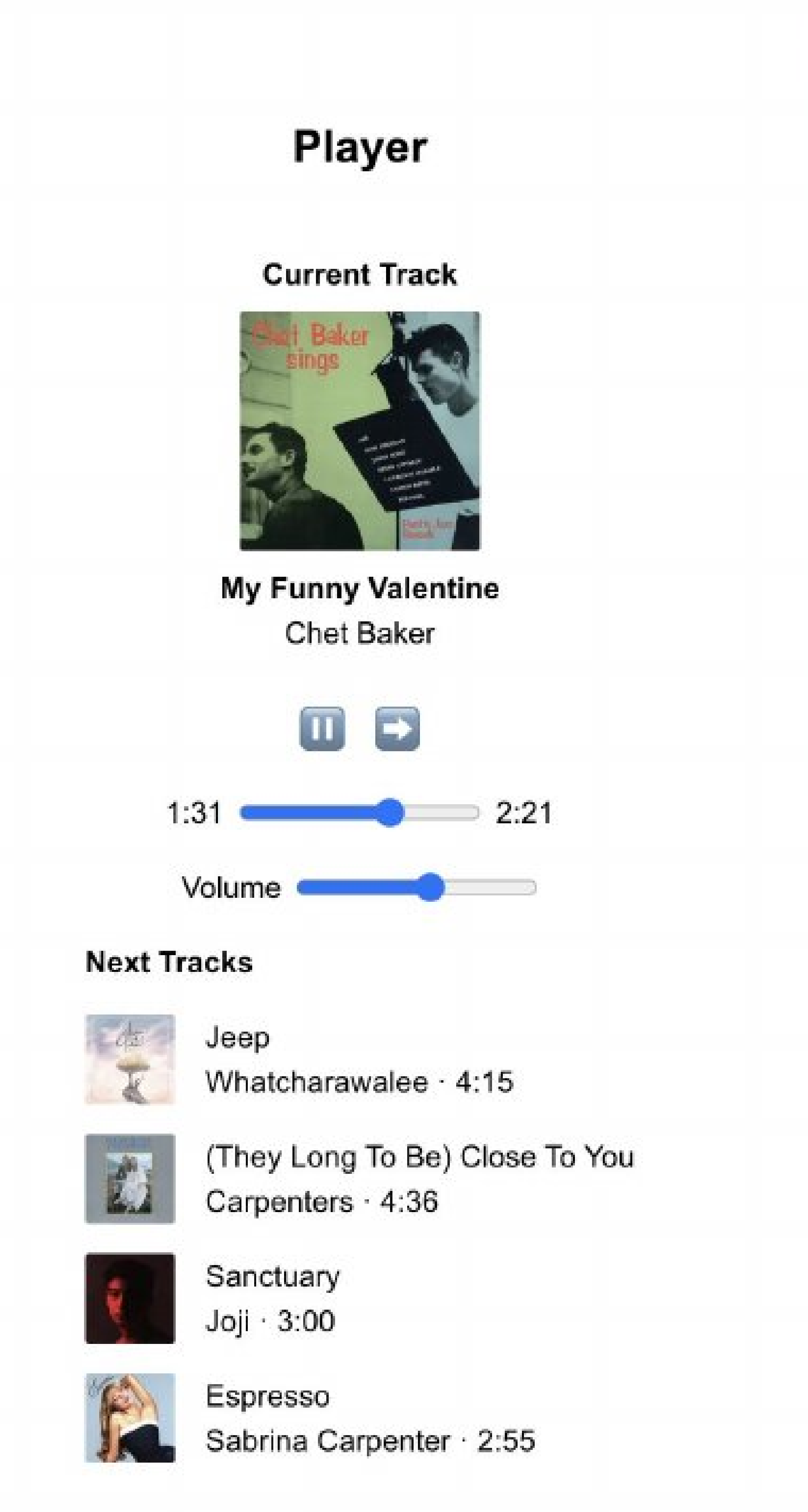


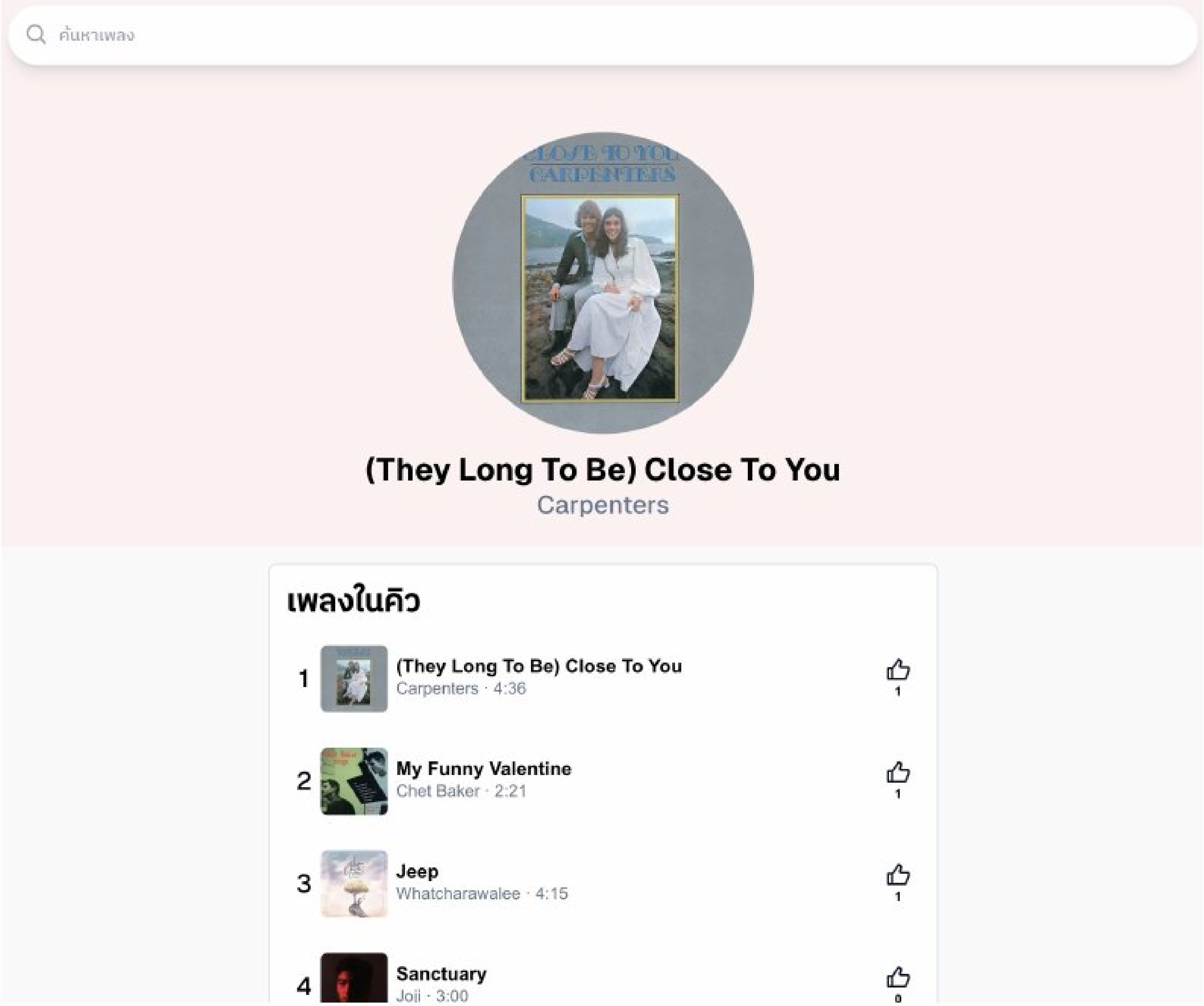




Raspberry Pi Pico Stepper Controller PCB

As a class project, I had the design and fabricate a Printed Circuit Board for controlling stepper motors using Raspberry Pi Pico. The designing process was done using EasyEDA.



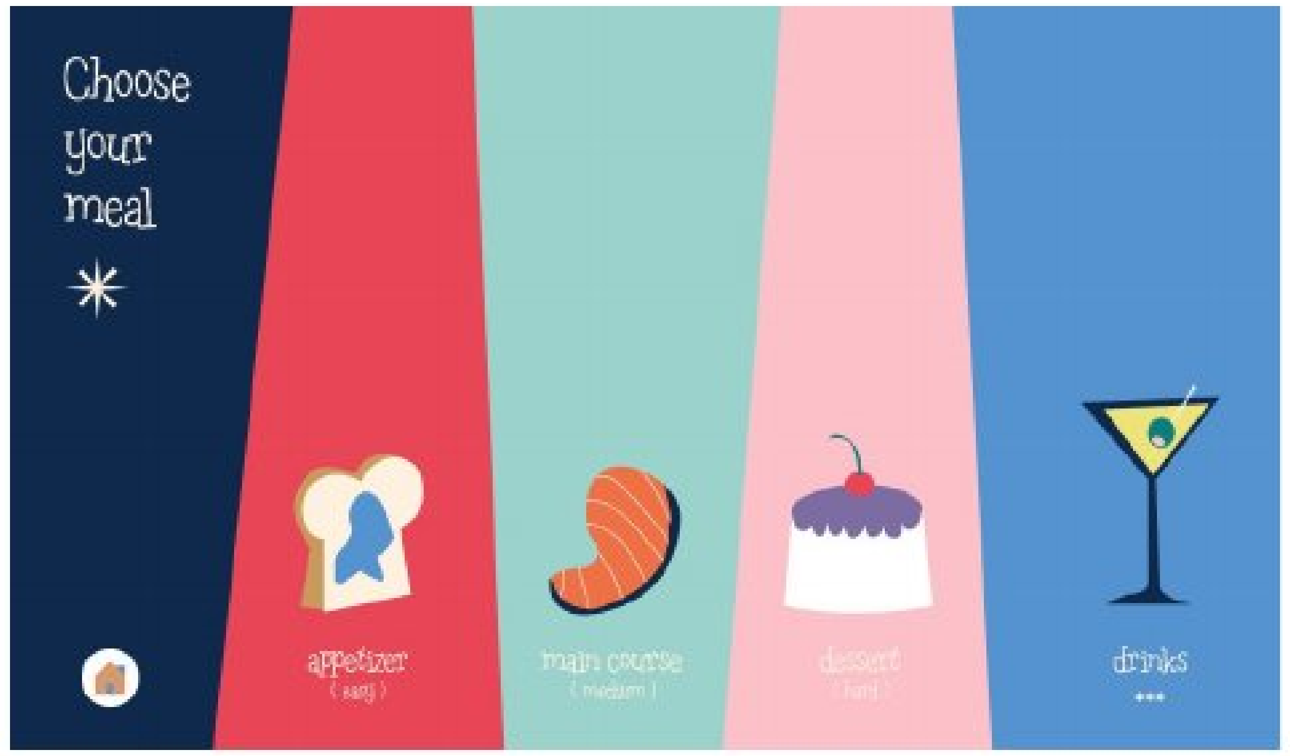


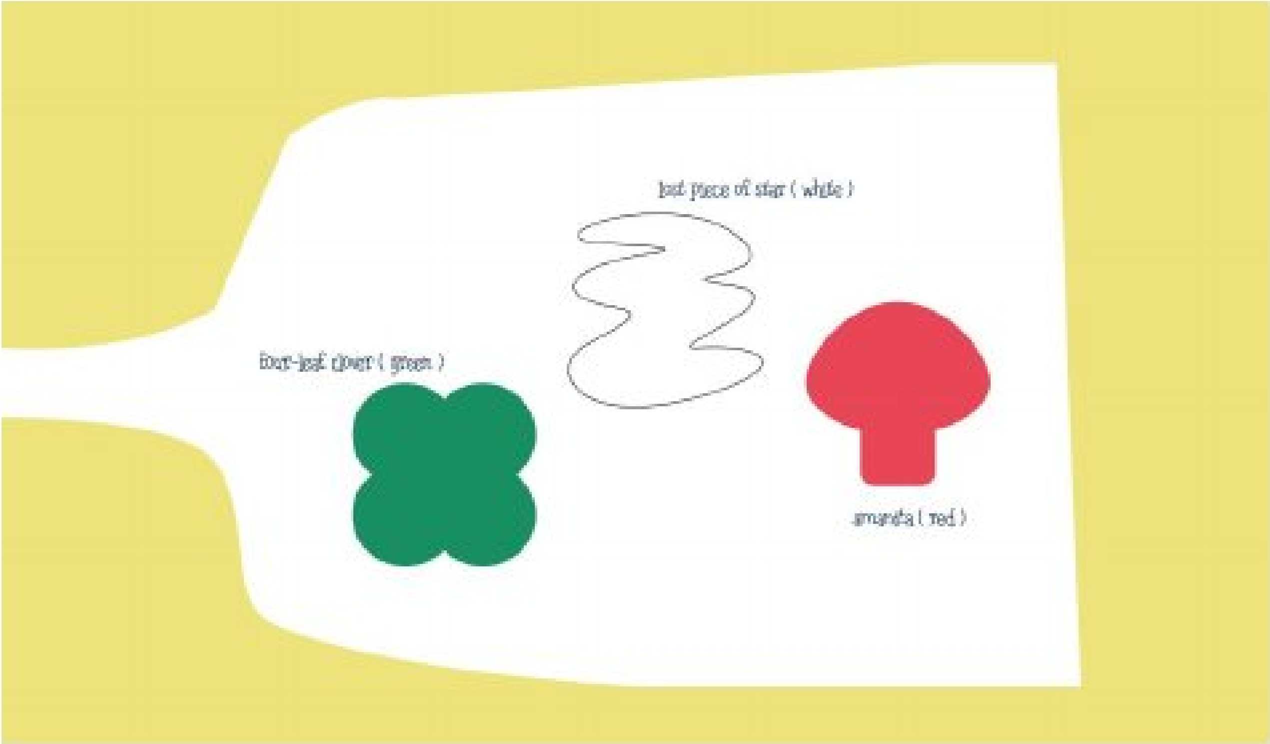
Raspberry Pi Music Player

GitHub Link

Role: Engineer

This is a music player controlled by an LED Keypad powered by a Raspberry Pi. Music is streamed and fetched from the Spotify API and includes volume control, playback control (pause, skip, jump), along with social features like “Liking” a song.





Color Guesser Toy

GitHub Link

Role: Engineer

The Color Guesser Toy is a toy where players place color blocks according to the prompt in the screen. It featured a display screen and camera module to filter for color, which is powered by a computer inside the casing. The front-end is built using ReactJS.

01 Robotics and Embedded Hardware

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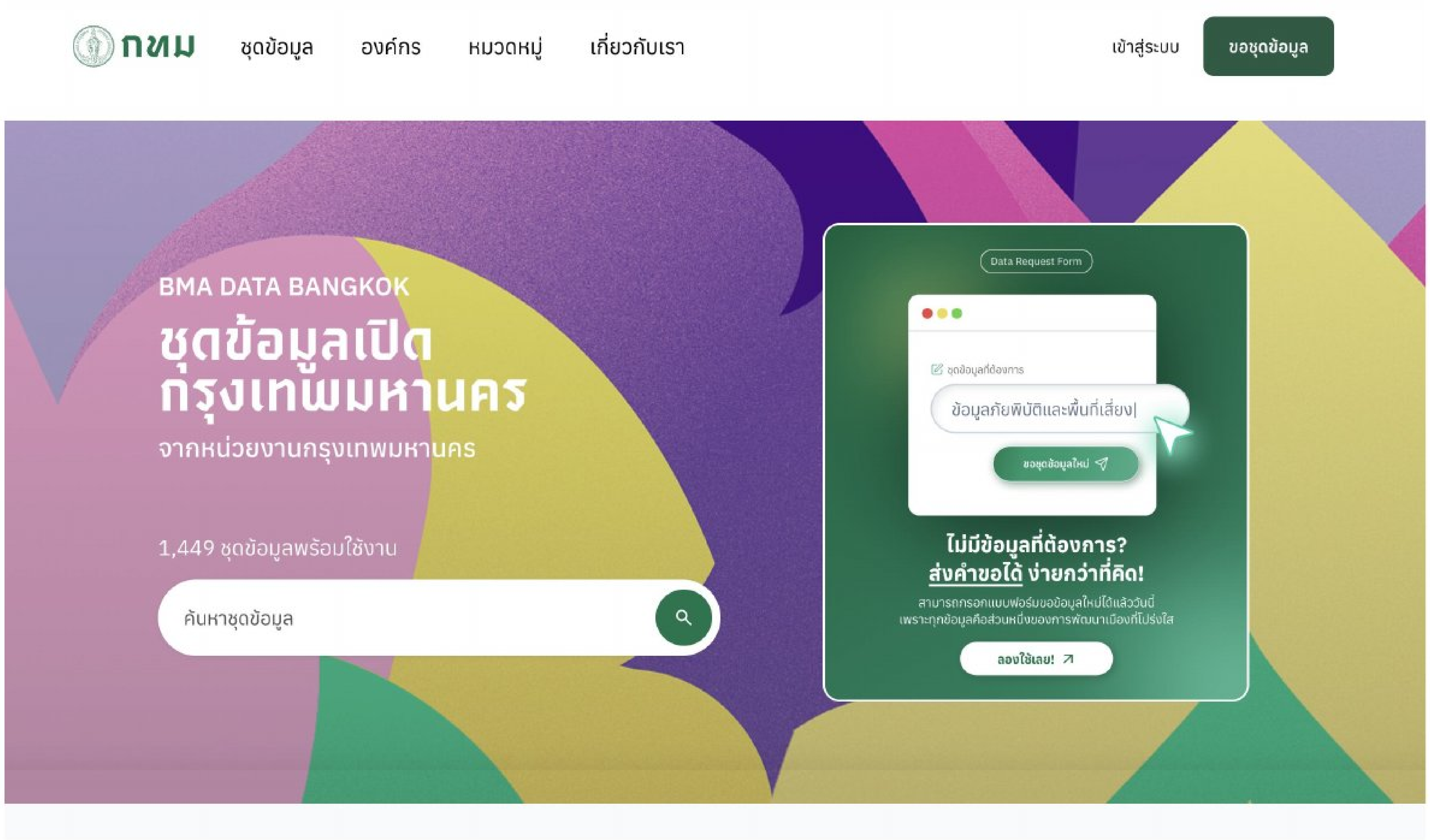
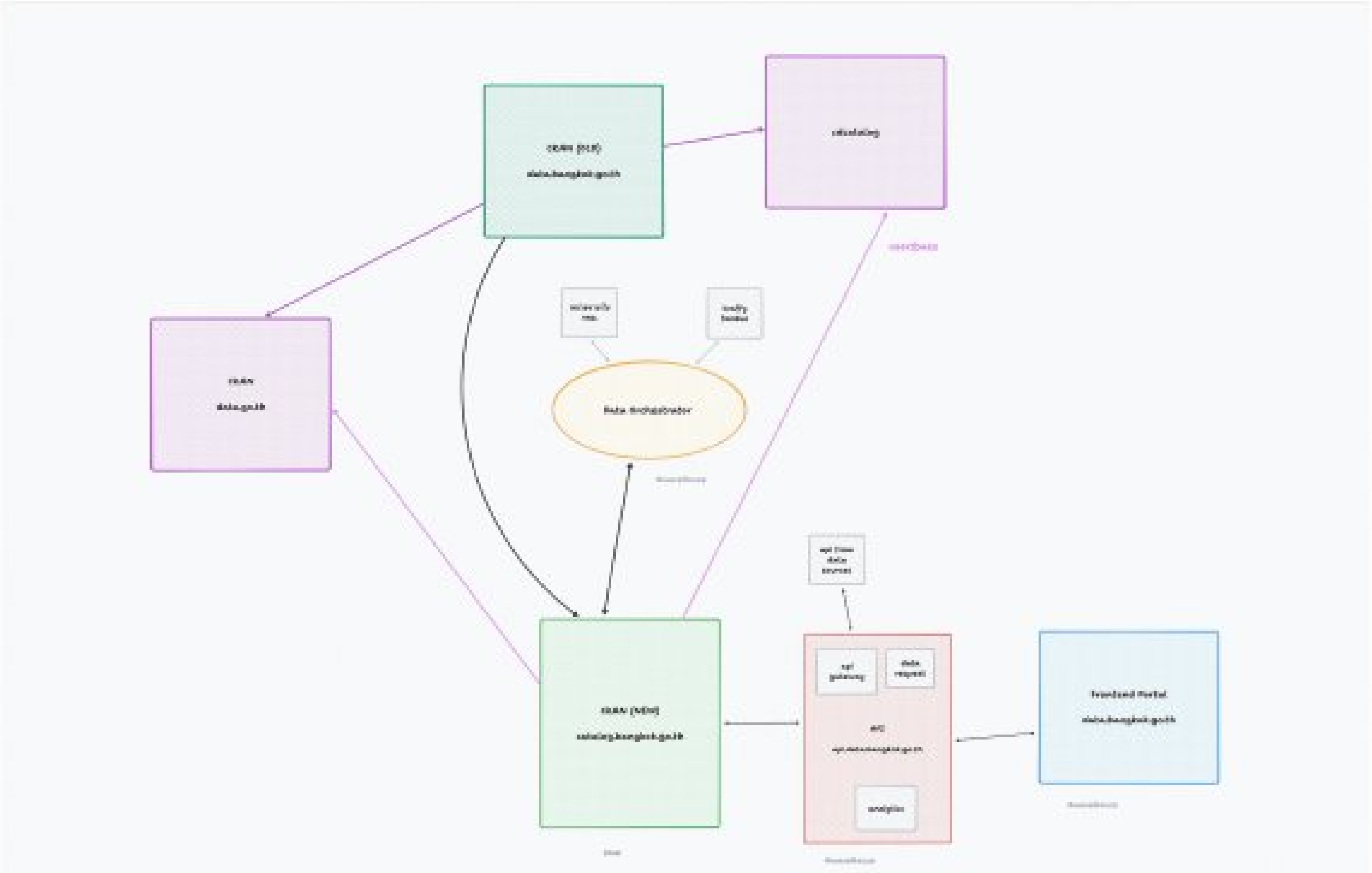
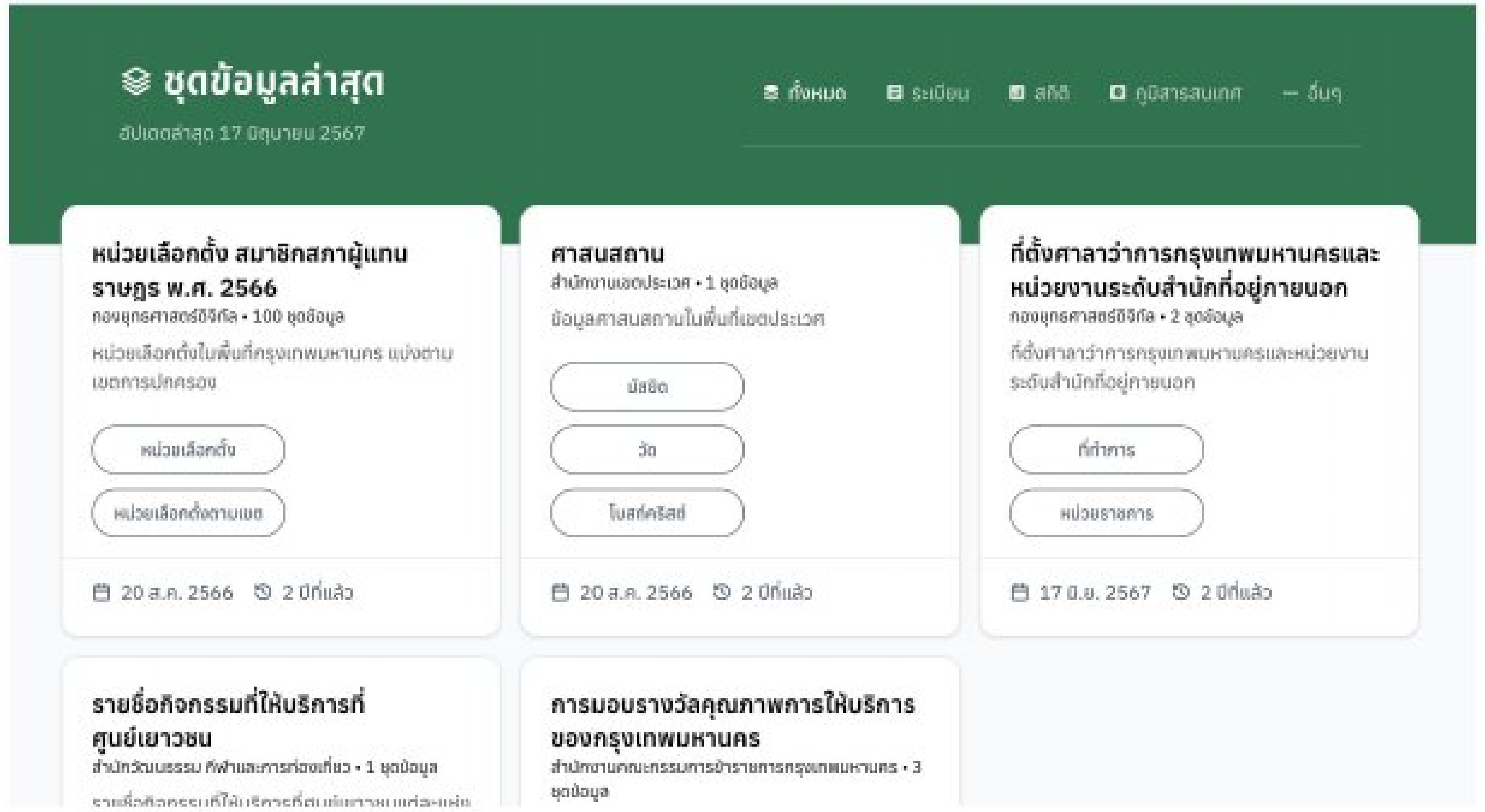




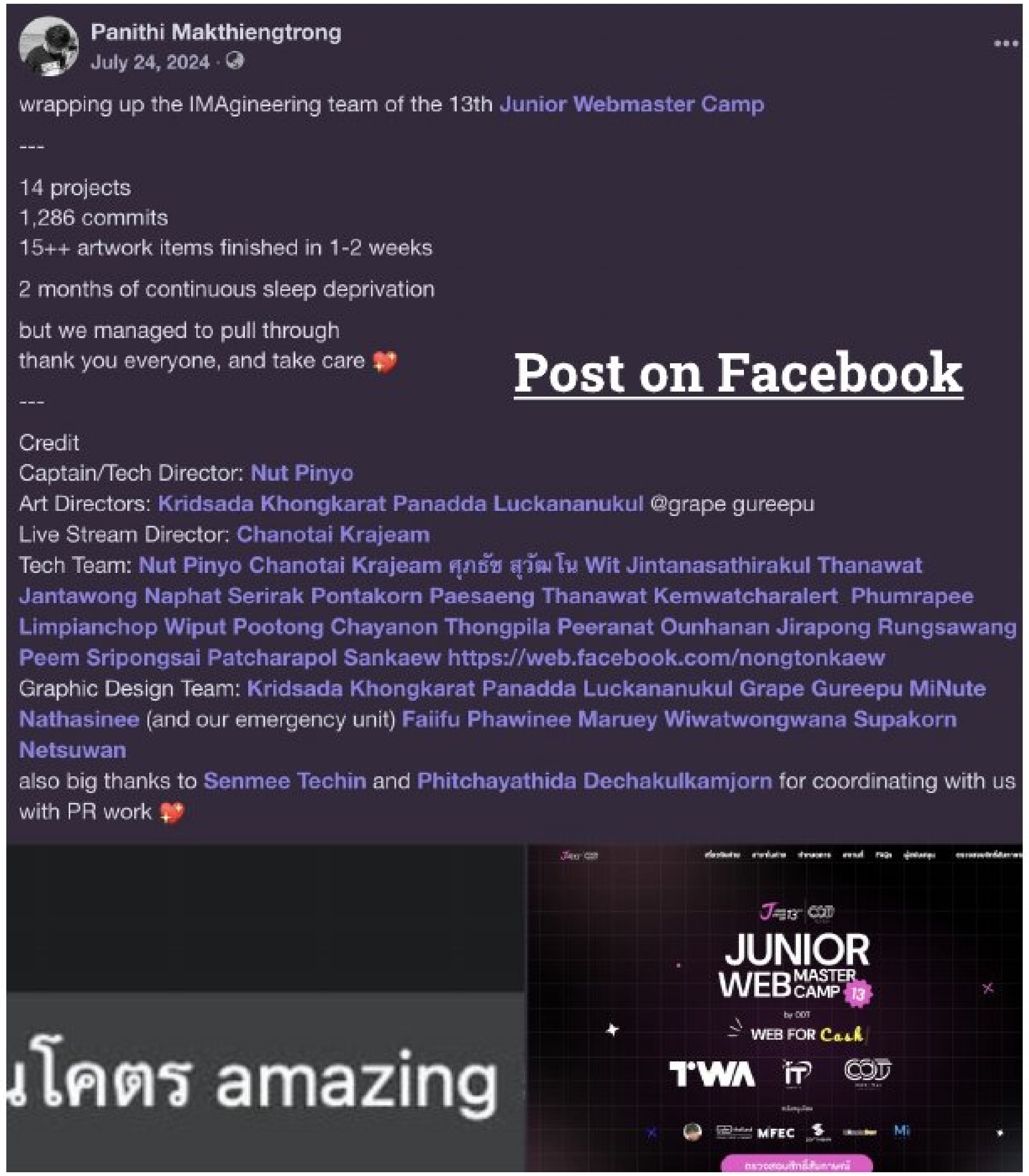
first architectural plan



Bangkok Open Data Platform

Role: Technical Lead

I led the technical development of the wrapper website for data.bangkok.go.th - an open data portal for Bangkok Metropolitan Administration's Open Government initiative - the process involved designing a static site building flow which optimized page load times in comparison to the old website. It fetched from a software called CKAN and acted as its much more performant wrapper.



Panithi Makthiengtrong
July 24, 2024

wrapping up the IMAgineering team of the 13th Junior Webmaster Camp

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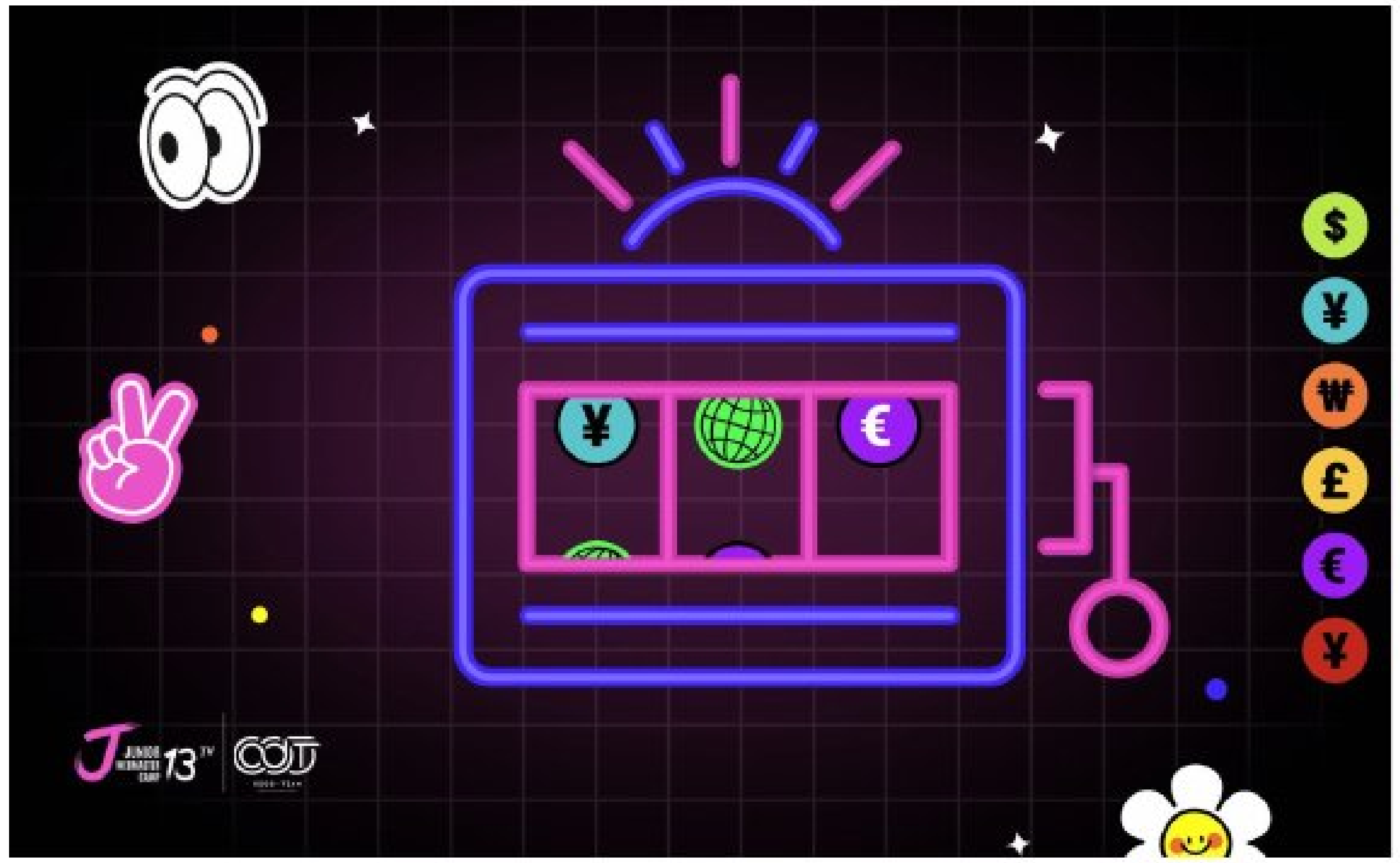
14 projects
1,286 commits
15++ artwork items finished in 1-2 weeks

2 months of continuous sleep deprivation

but we managed to pull through
thank you everyone, and take care 💖

Credit
Captain/Tech Director: Nut Pinyo
Art Directors: Kridsada Khongkarat Panadda Luckananukul @grape gureepu
Live Stream Director: Chanotai Krajeam
Tech Team: Nut Pinyo Chanotai Krajeam ศุภธัช สุวัฒโน Wit Jintanasathirakul Thanawat Jantawong Naphat Serirak Pontakorn Paesaeng Thanawat Kemwatcharalert Phumrapee Limpianchop Wiput Pootong Chayanon Thongpila Peeranat Ounhanan Jirapong Rungsawang Peem Sripongsai Patcharapol Sankaew https://web.facebook.com/nongtonkaew
Graphic Design Team: Kridsada Khongkarat Panadda Luckananukul Grape Gureepu MiNute Nathasinee (and our emergency unit) Faiifu Phawinee Maruey Wiwatwongwana Supakorn Netsuwan
also big thanks to Senmee Techin and Phitchayathida Dechakulkamjorn for coordinating with us with PR work 💖

Post on Facebook



Junior Webmaster Camp 13

Role: Director of Sponsorship Team (JPEX), Interim Director of Imagineering Team (IMA)

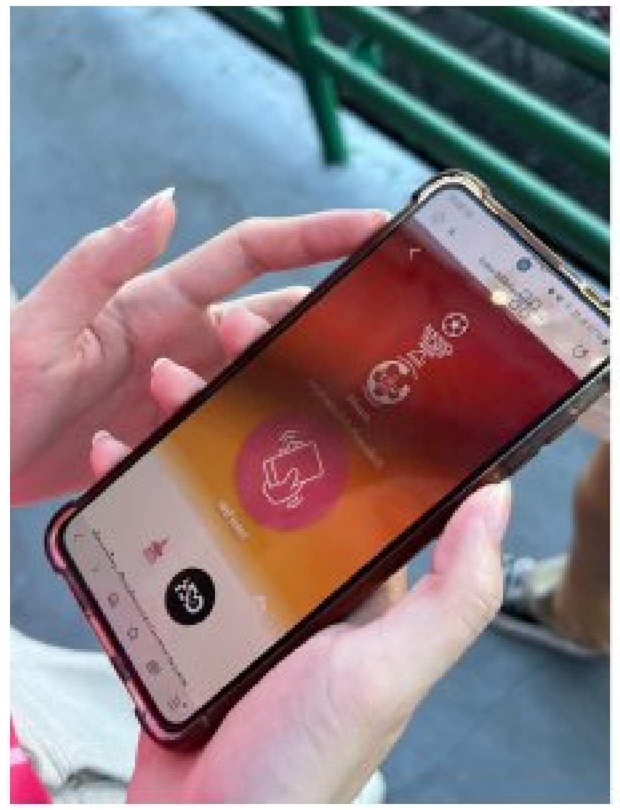
I acted as Interim Director of the Imagineering Department of the 13th Junior Webmaster Camp.
During the time-intensive development timeline, I managed the team in creating of event decoration elements, swag, and the creation of multiple websites used throughout the event.

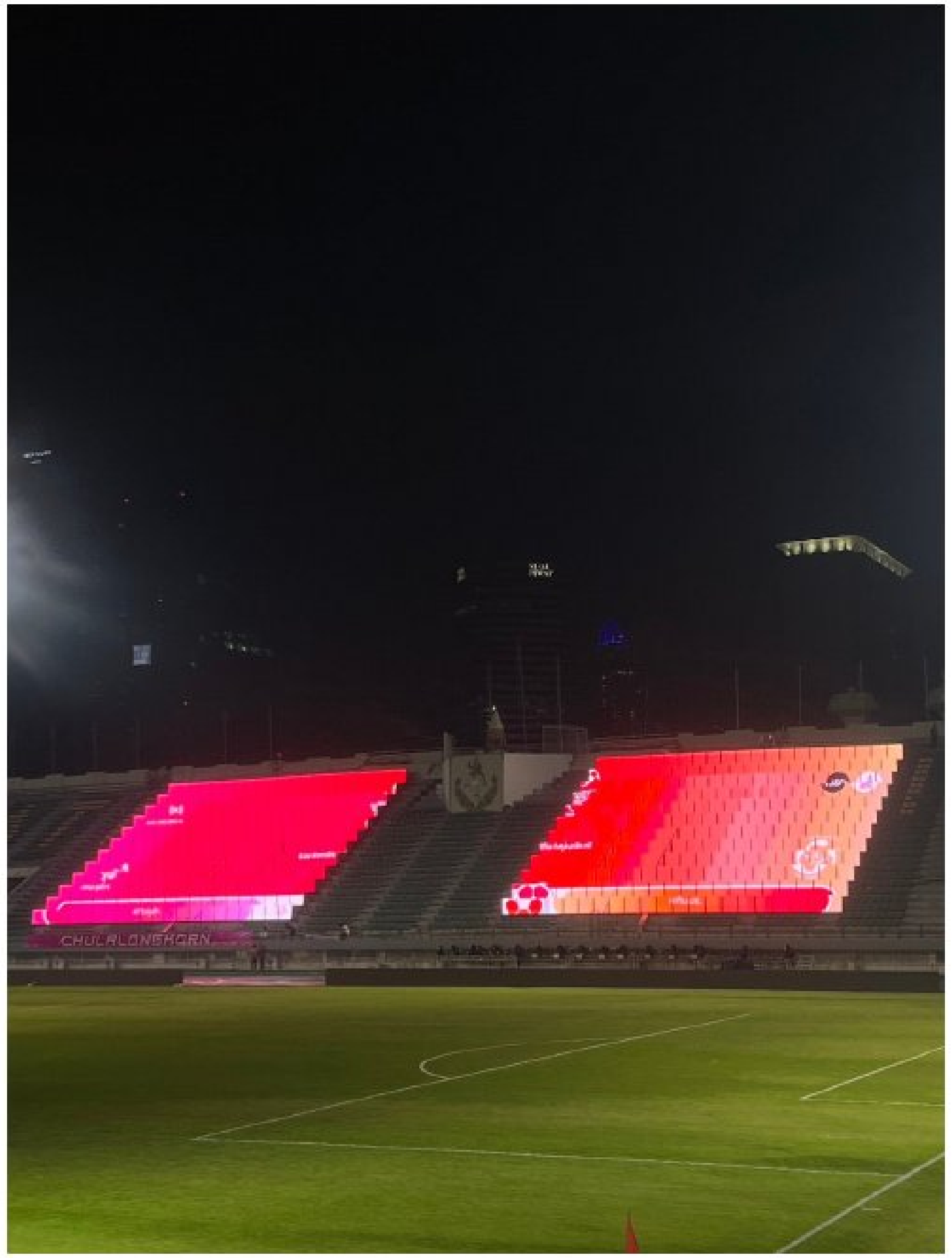
ISD SGCU

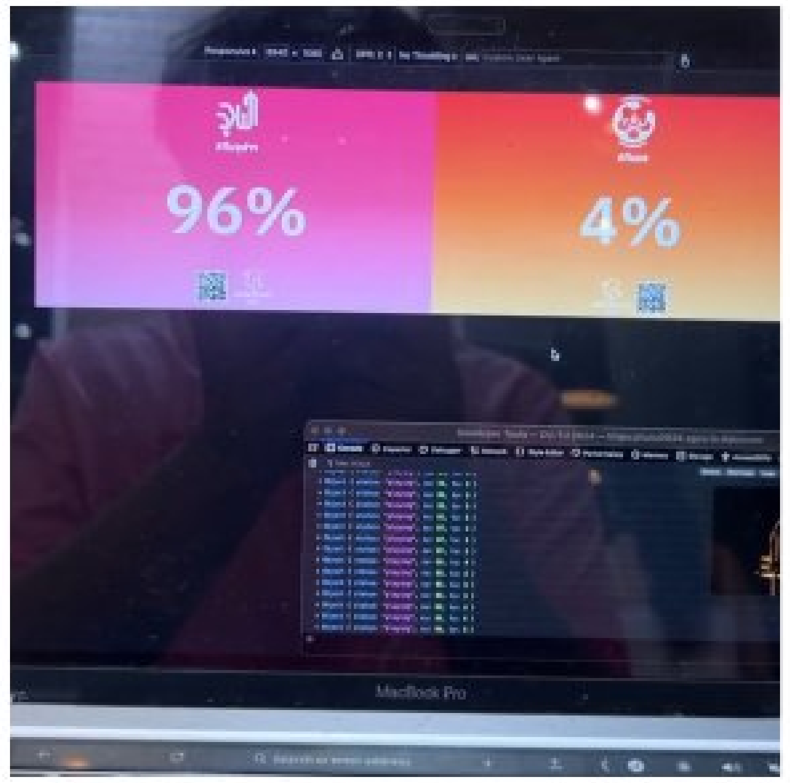
Role: Department President

I acted as department president for the Information System Development department of the Student Government of Chulalongkorn University during 2023-24.

Managed the 30+ members team that are responsible for making digital systems used by more than 6,000 users in Chulalongkorn University brought me valuable experience on team and project management in a large scale.







CU-TU Unity Ball 2024

Role: Project Manager

Our team was faced with a challenge — we had 2 weeks to develop an interactive pre-match game for participants to play at the CU-TU Unity Ball Match Event.

We created a game where people had to shake for either CU or TU's side and see who gets the most score.

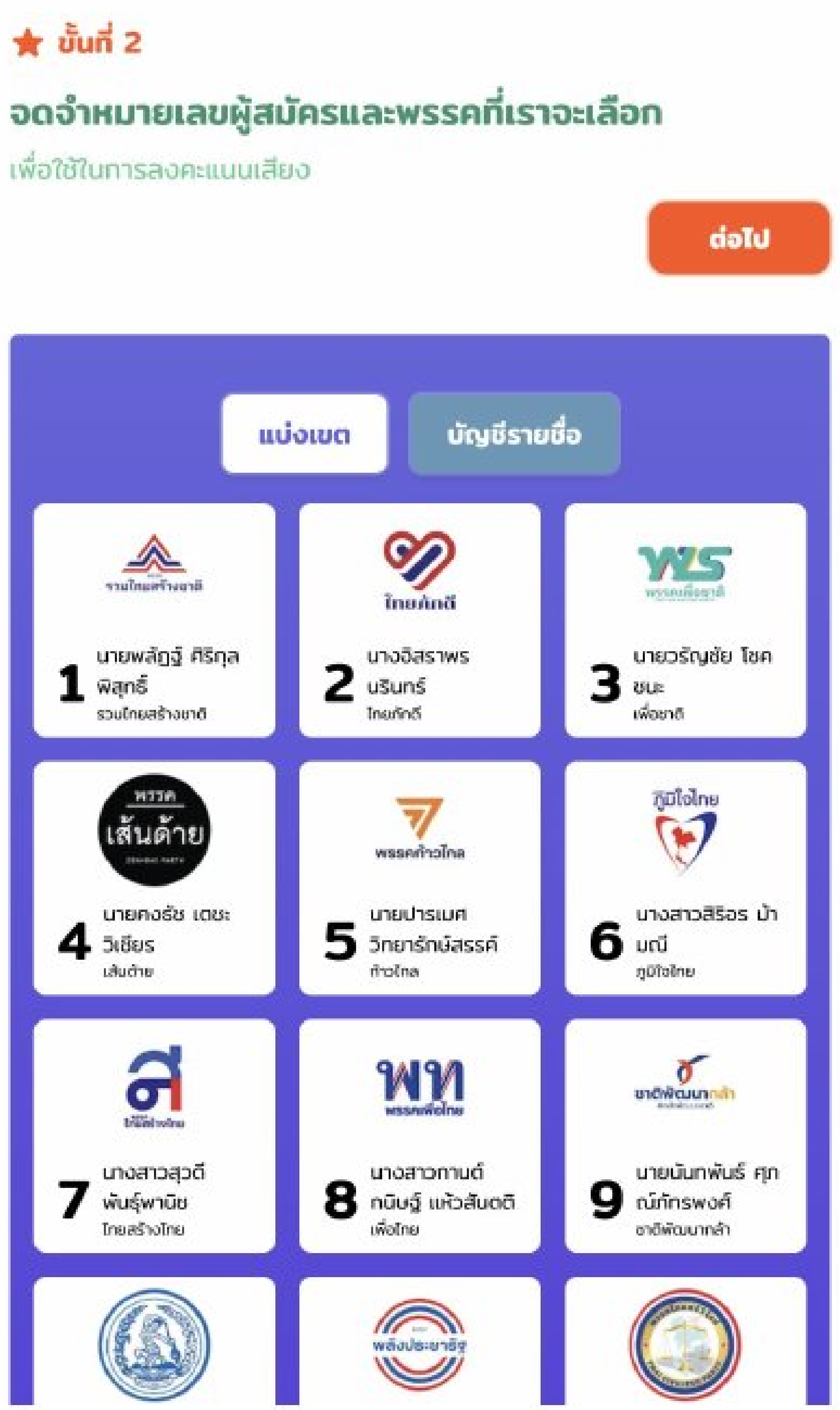
Our team designed the interface to meet the unusual scale of the LED screens from the stands, and the output is a quite-amusing event that is posted on TikTok and other platforms.

Rub Puen Kao Mai

Role: Project Supervisor, Web Developer

Rub Puen Kao Mai, or the Freshmen welcome event is an event hosted in Chulalongkorn University to welcome new students.

I supervised and collaborated with stakeholders to create the RPKM website, a website that matches users with their preferred house (activity group) and included the E-Stamp feature — where participants can collect stamps from events to exchange for rewards. It also included an E-Ticket system that identifies and tracked participants of the Freshmen Night concert where 3,000+ people had attended the event.



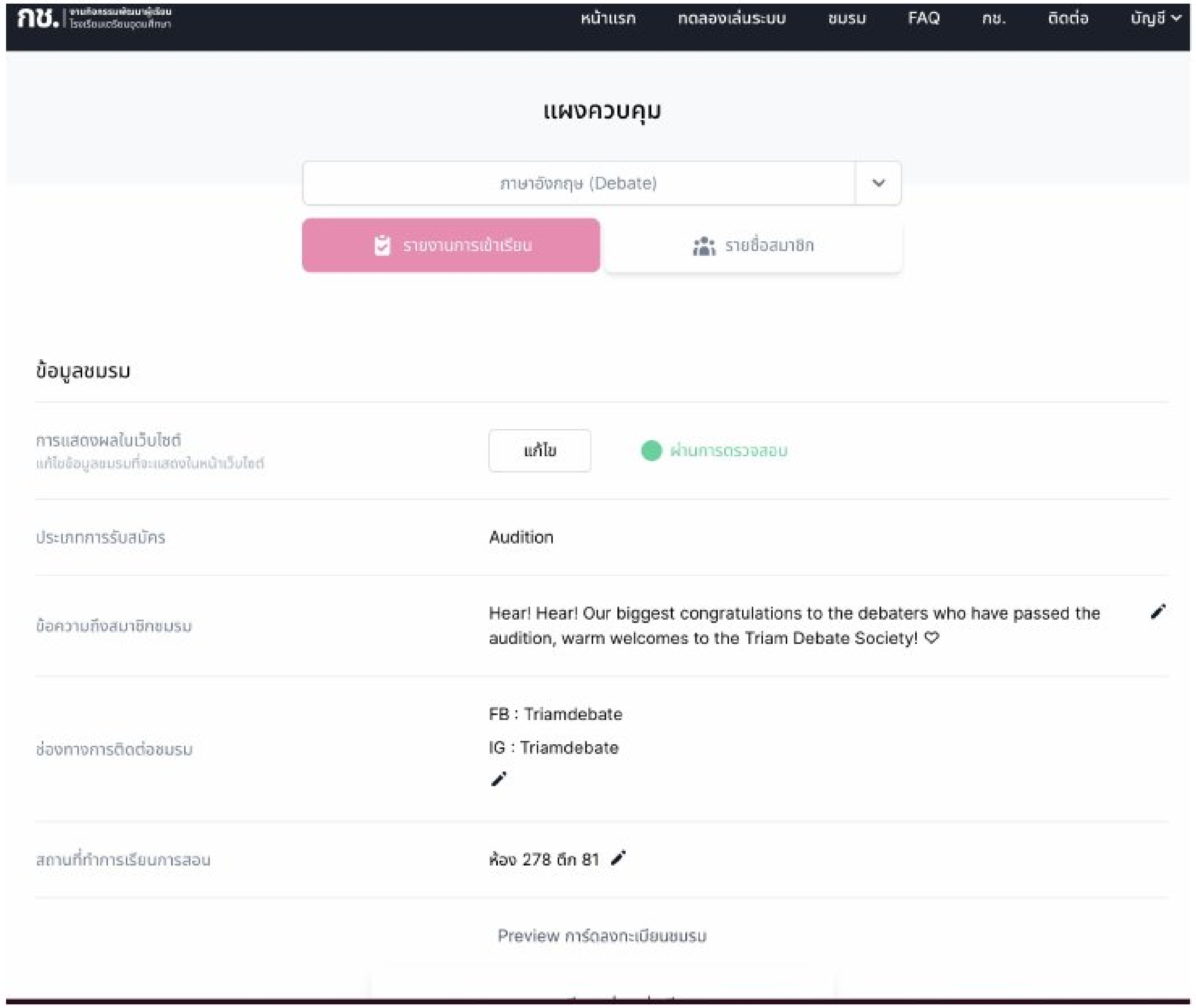
GitHub Link

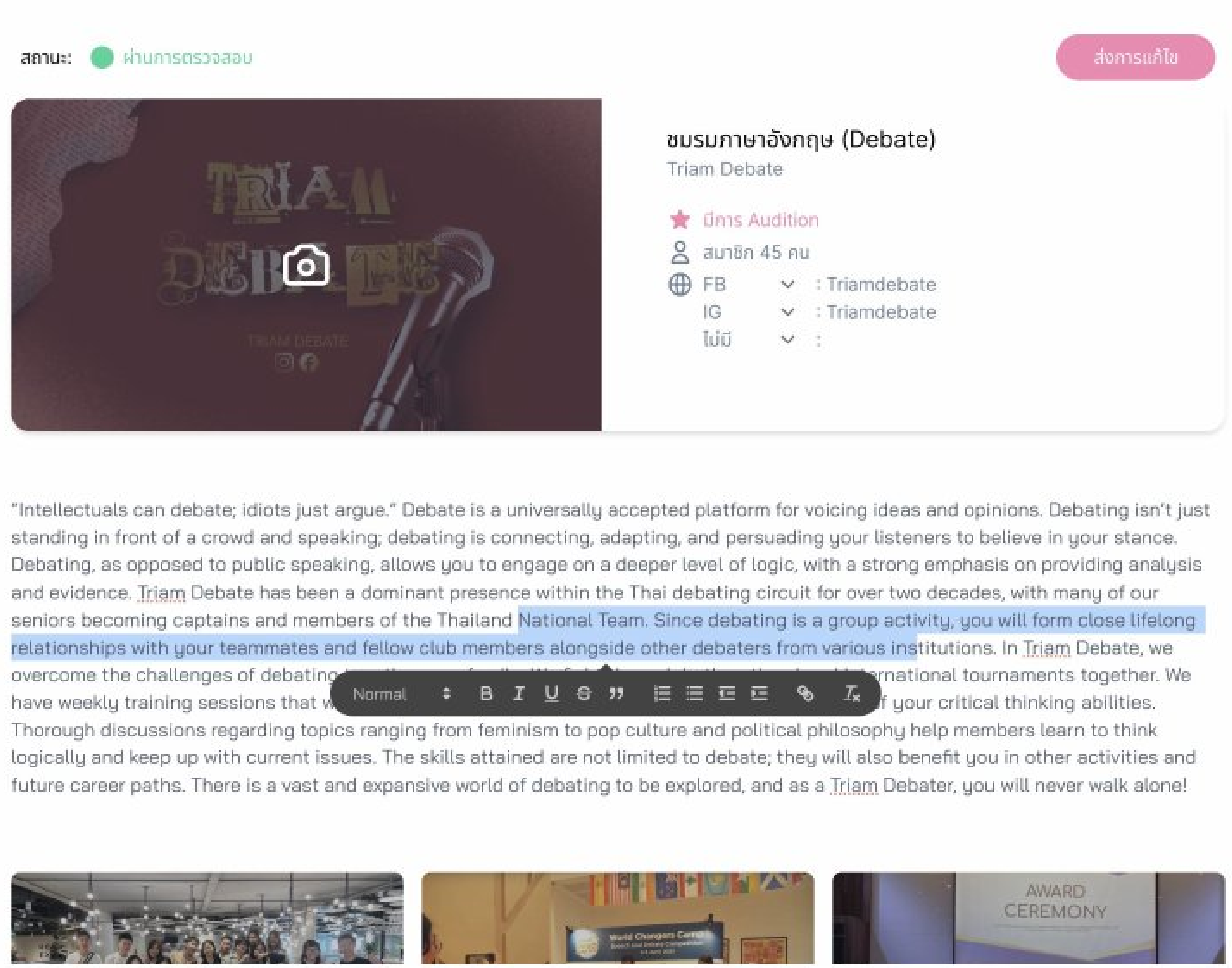
PED4YOU (Election Simulator)

Role: Web Developer

I was the project coordinator for the PED4YOU project an election simulator and information center that me and my first-voter friends and I built for the 2023 Thai general election.

My role involved building the website and coordinating with my ML developer and UX/UI Designer friend for building the election simulator project — a project that guides the user through steps on the processes of voting, with an infographic and an AI ballot verifier feature, which is intended on informing first voters on how to vote.





Triam Udom Club Registration System

Role: Web Developer

During 2021-22, I was the web developer of Traim Udom Clubs Management Committee. During the time, I maintained and built the club registration that is used by every student in the school. Features include a club auditioning system, content management system, and a registration simulator feature.

Triam Udom Online Loy Kratong Website

Role: Web Developer

Me and the TUCMC team built an online loy kratong website for our high school during the COVID-19 pandemic. It featured customizable and draggable kratongs, custom messages, and savable images for sharing on social media.

The website was built using Next.js and the database is on Firestore.

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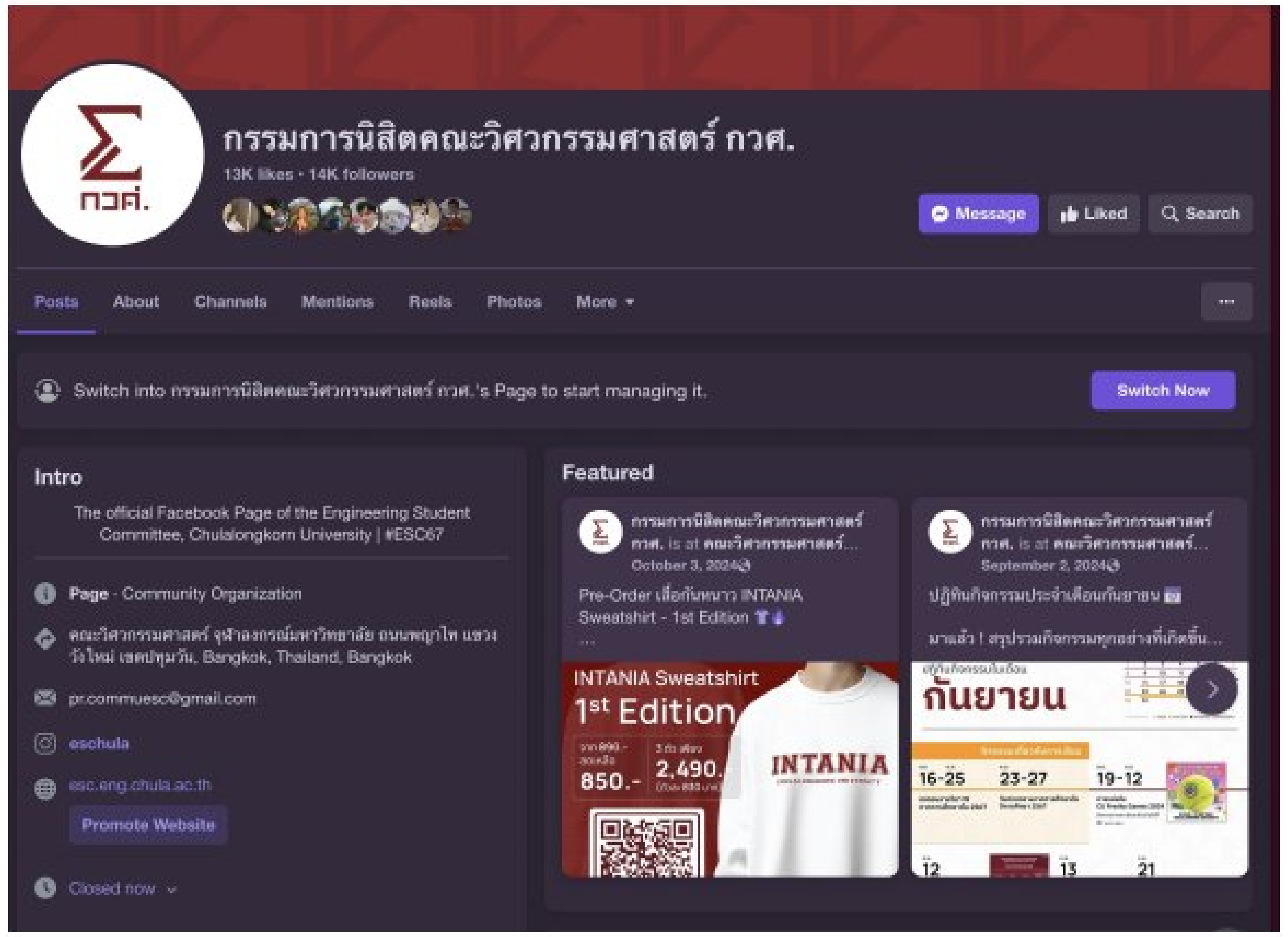
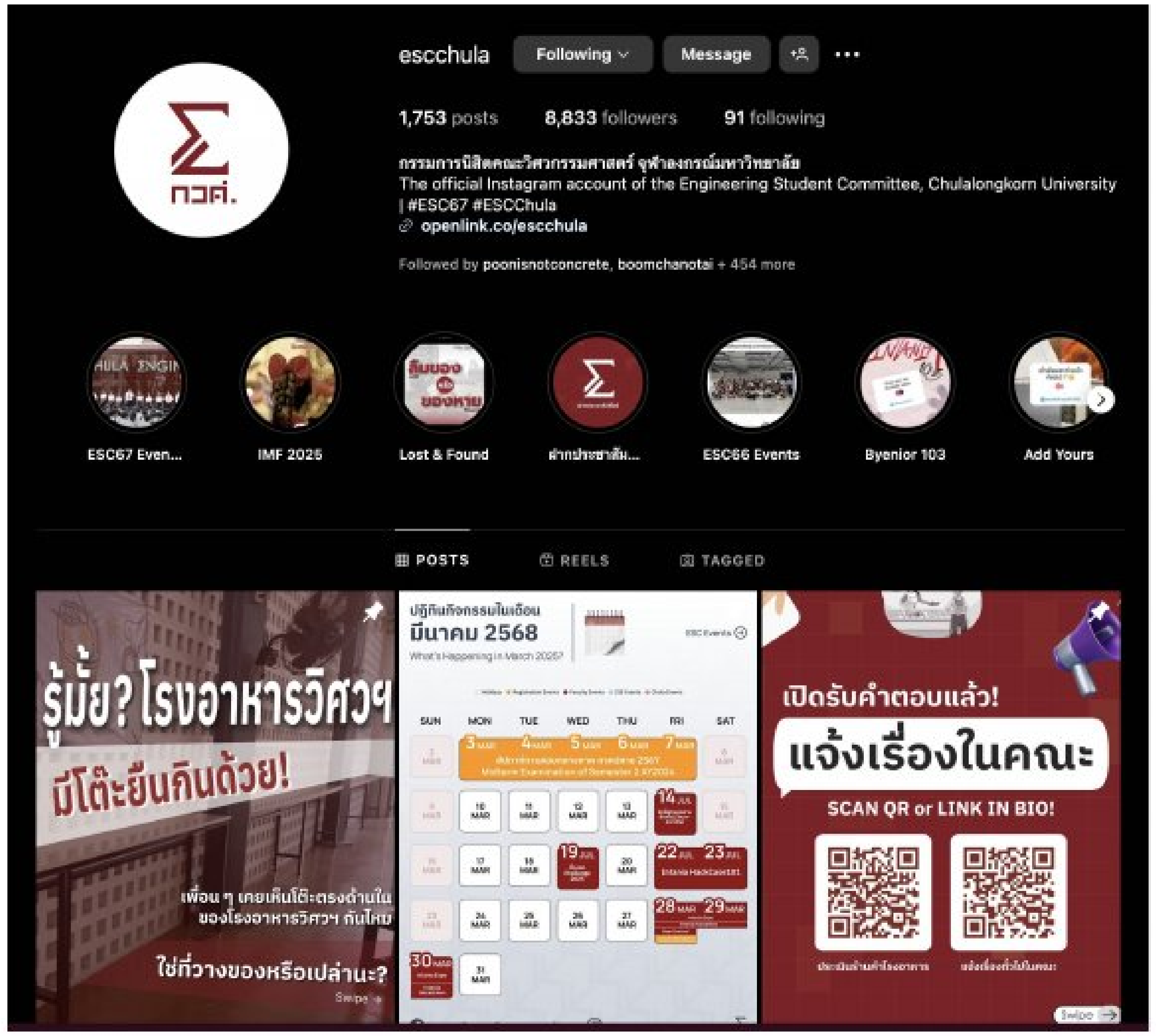
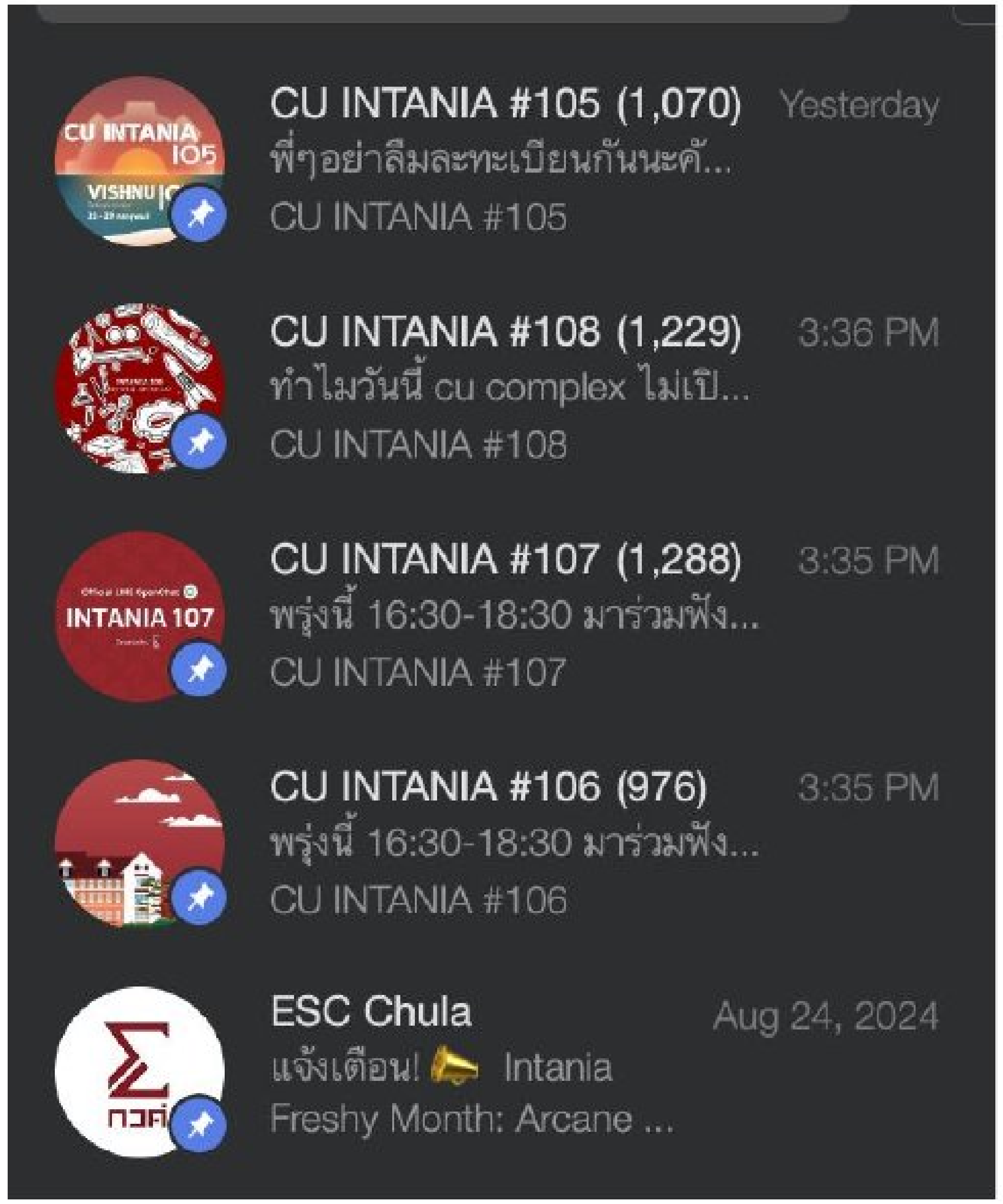






Engineering Student Committee (ESC Chula)

Role: MARCOM Department Lead

Engineering Student Committee (ESC) is a student-run body containing over 100 people who work on events and projects that benefit the people of engineering faculty, Chulalongkorn University.

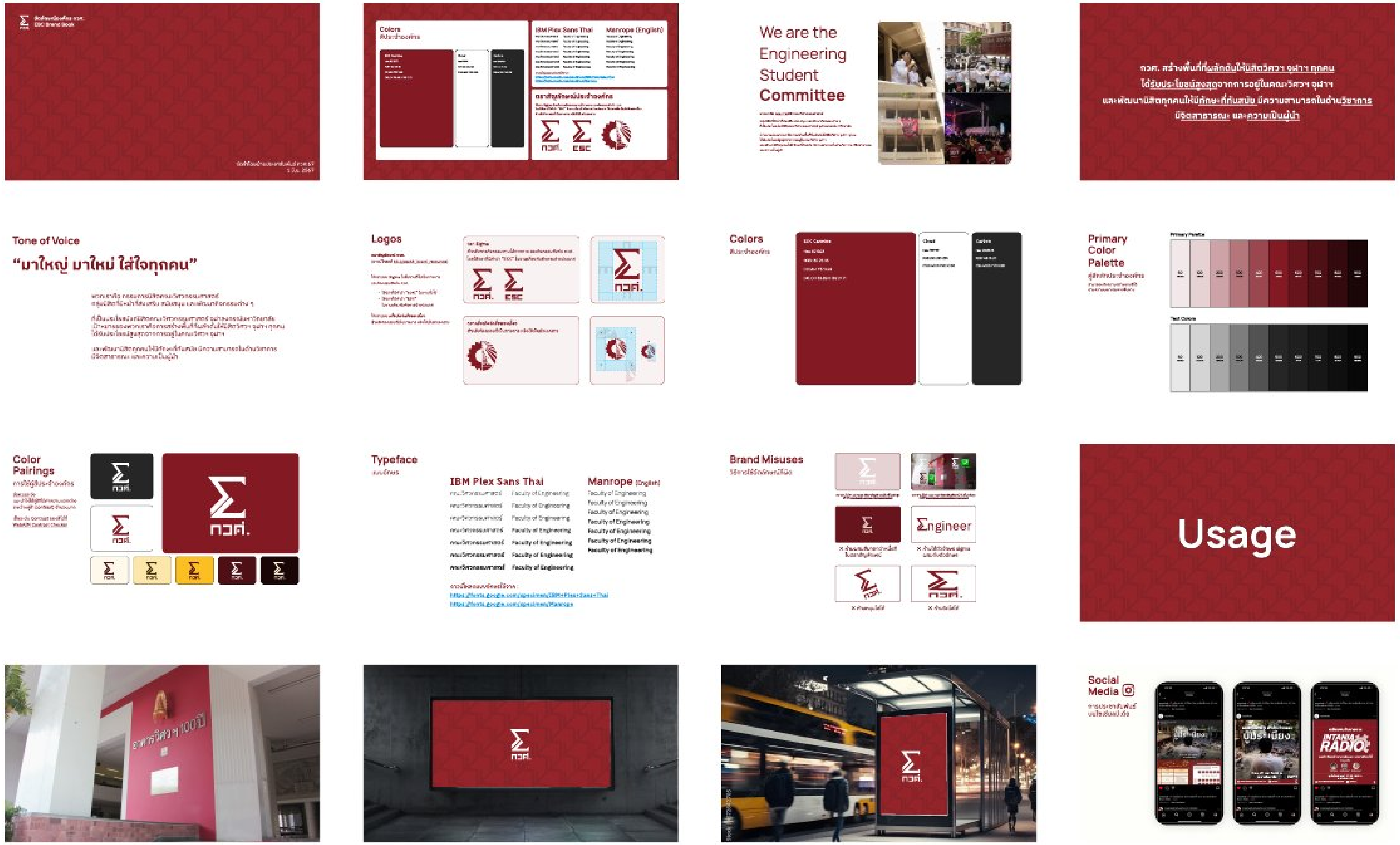
I managed Instagram, Facebook, and LINE OpenChat channels for the Engineering Student Committee, Chulalongkorn University, reaching an audience of 5,000+ members. This responsibility strengthened my awareness of real-world impact and sharpened my approach to user-centric design.

ESC Brand Identity

Role: Graphic Designer

During the start of my working term, I had the chance to refine the brand identity documentation of ESC Chula, which is mostly a definition of a style that had emerged and been lightly documented for a while.

During the research process, I had learned of how other companies present their Corporate Identities in their documentation and have served as inspiration for this project.

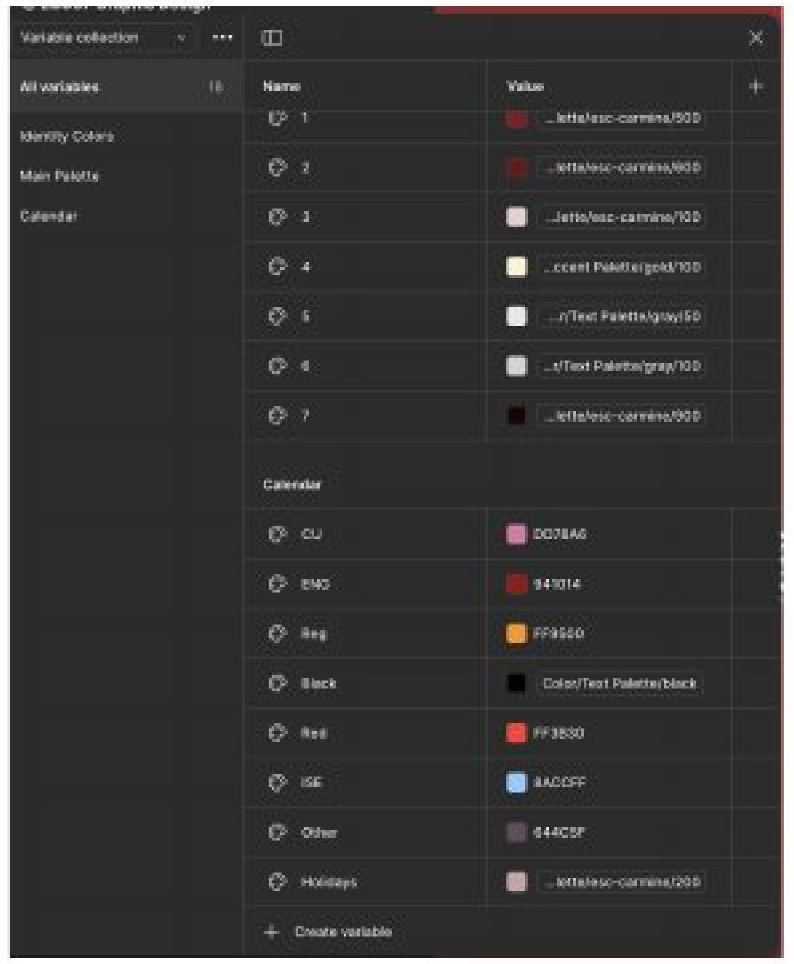


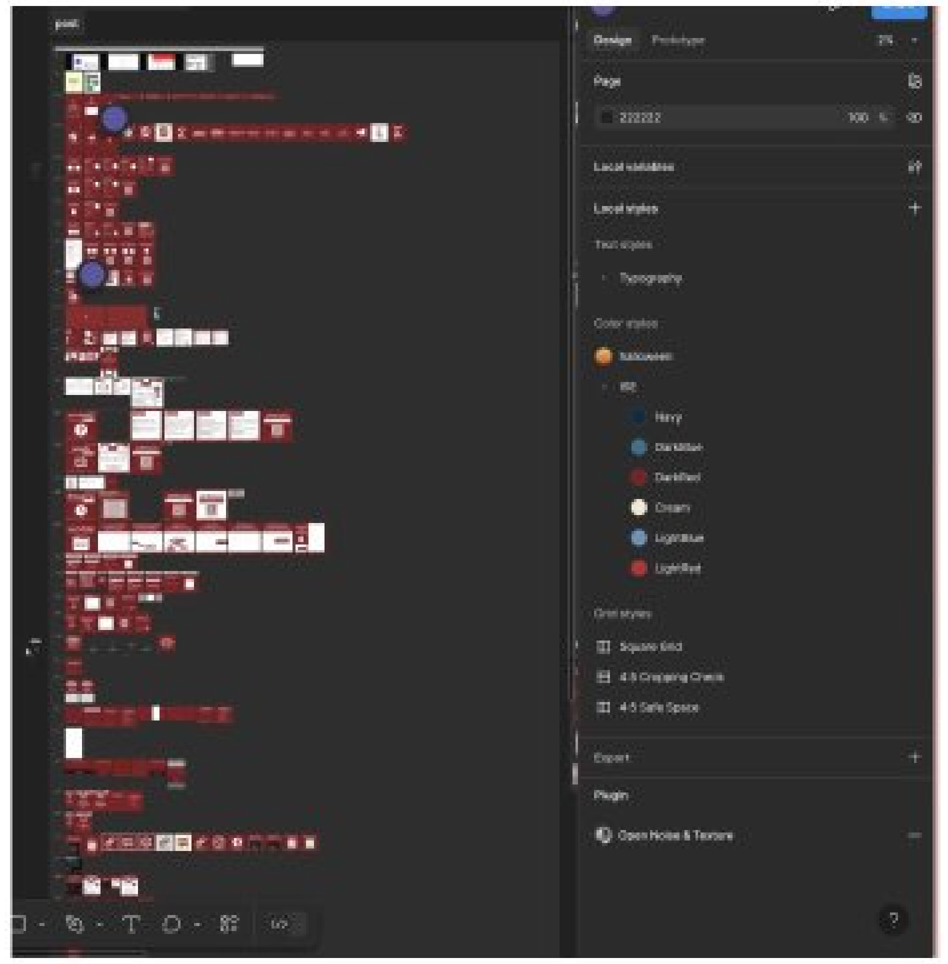
ESC Figma Design Tokens

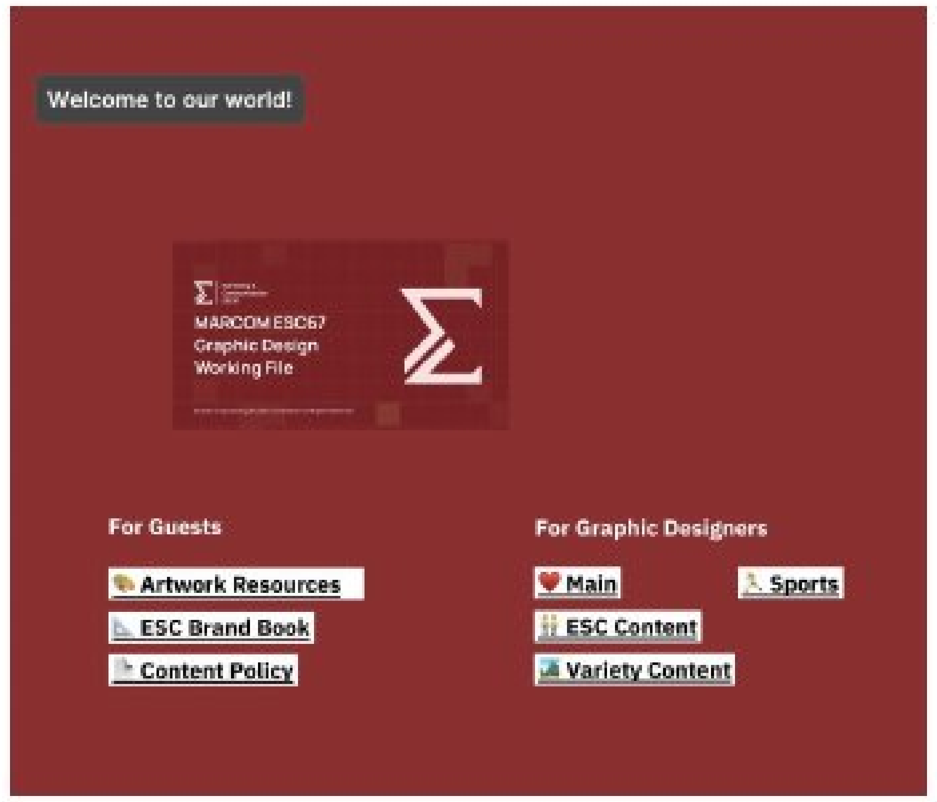
Role: Graphic Designer

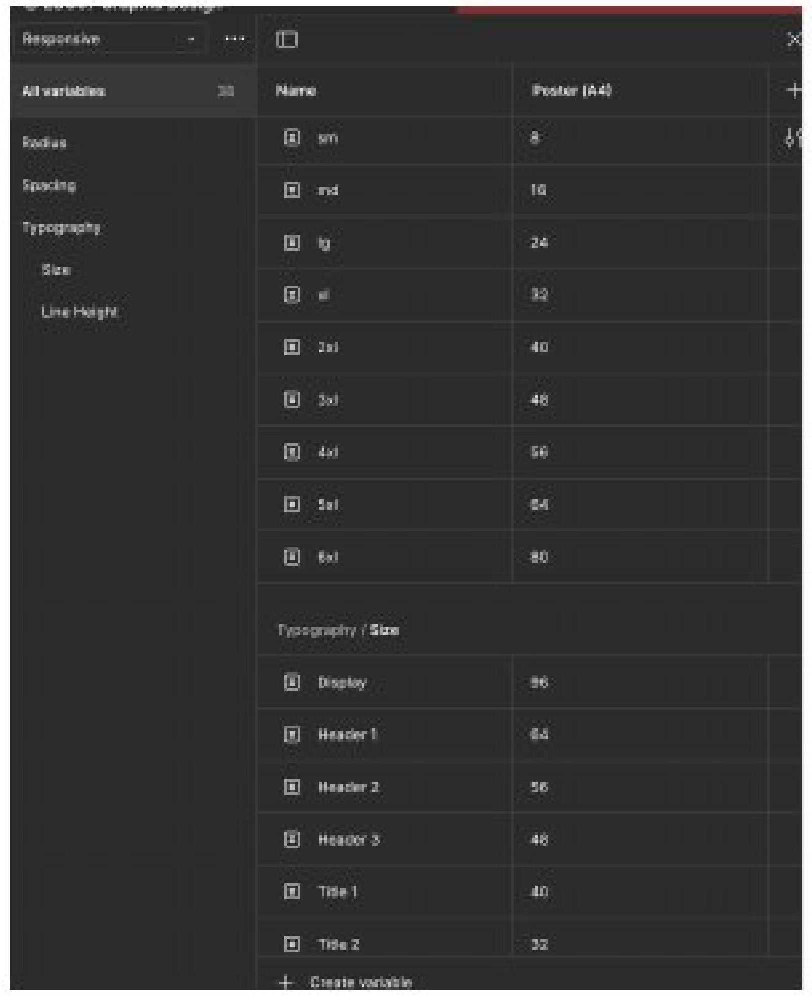
I set up a design system for producing posters and infographics using Figma.

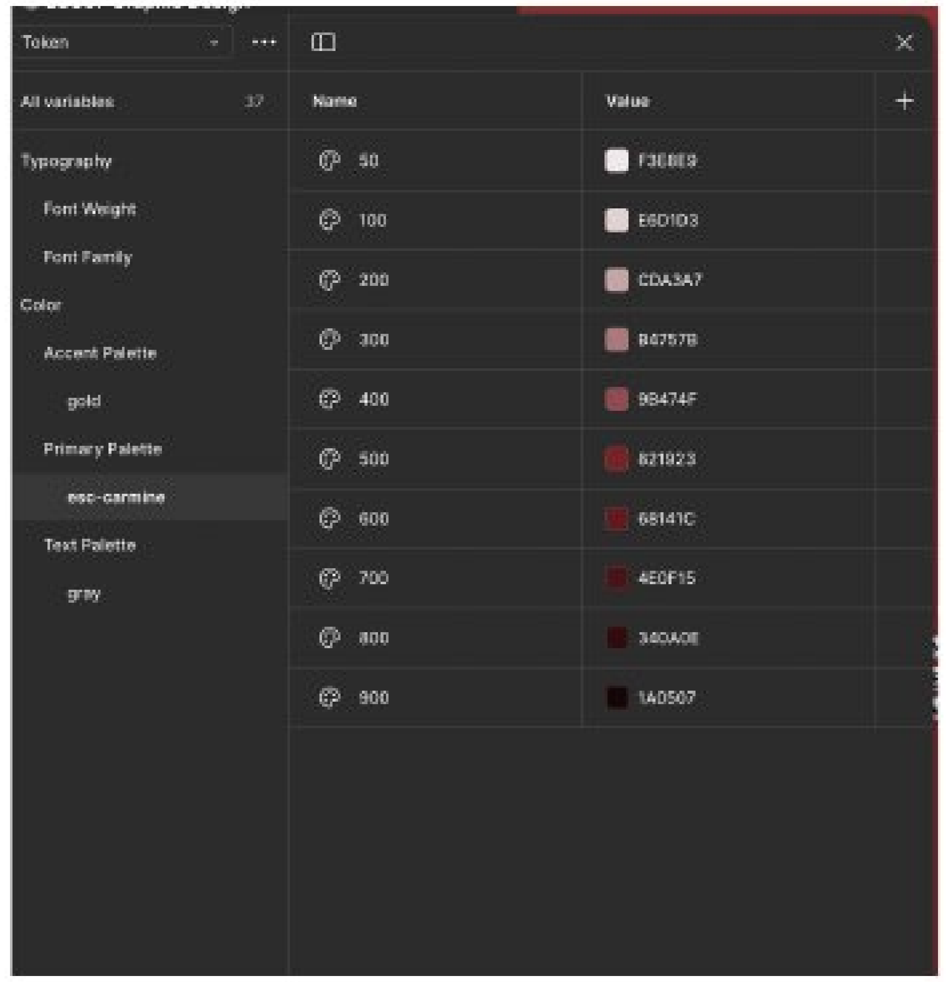
Setting up the variables and token system was really eye-opening for on the complex systems Figma have set up to improve productivity for its users, and it did really improve my design workflow significantly.

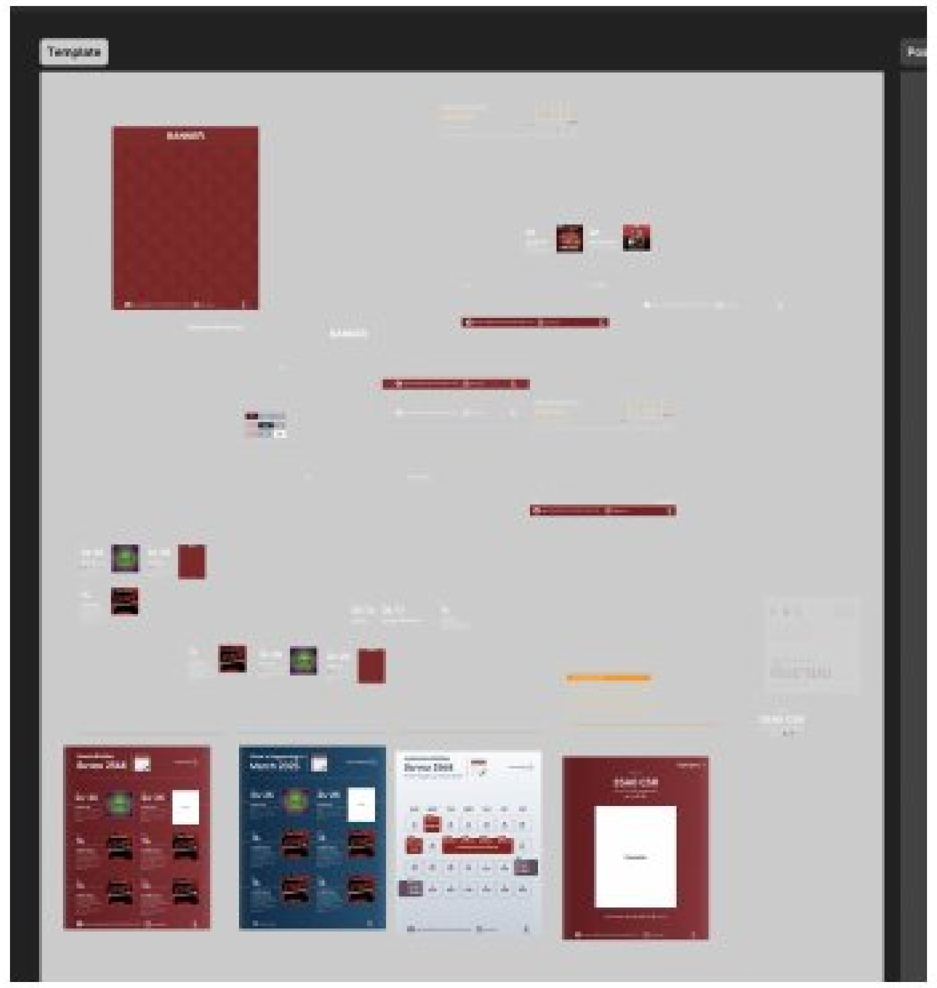












ESC Posters

Role: Graphic Designer

Me and my team created 300+ artwork for publicizing events in the Faculty of Engineering.

Here is a brief collection of my artwork for the Engineering Student Committee of Chulalongkorn University.

Intania Shop

Role: Project Lead

I pioneered the Intania Shop project, a place where student-made products from faculty of engineering can be sold to anyone with support from the faculty.

The launch process involved a daunting mix of business ideation, marketing communication, product development, accounting and finance and much more, which me and my team had to learn on the fly.

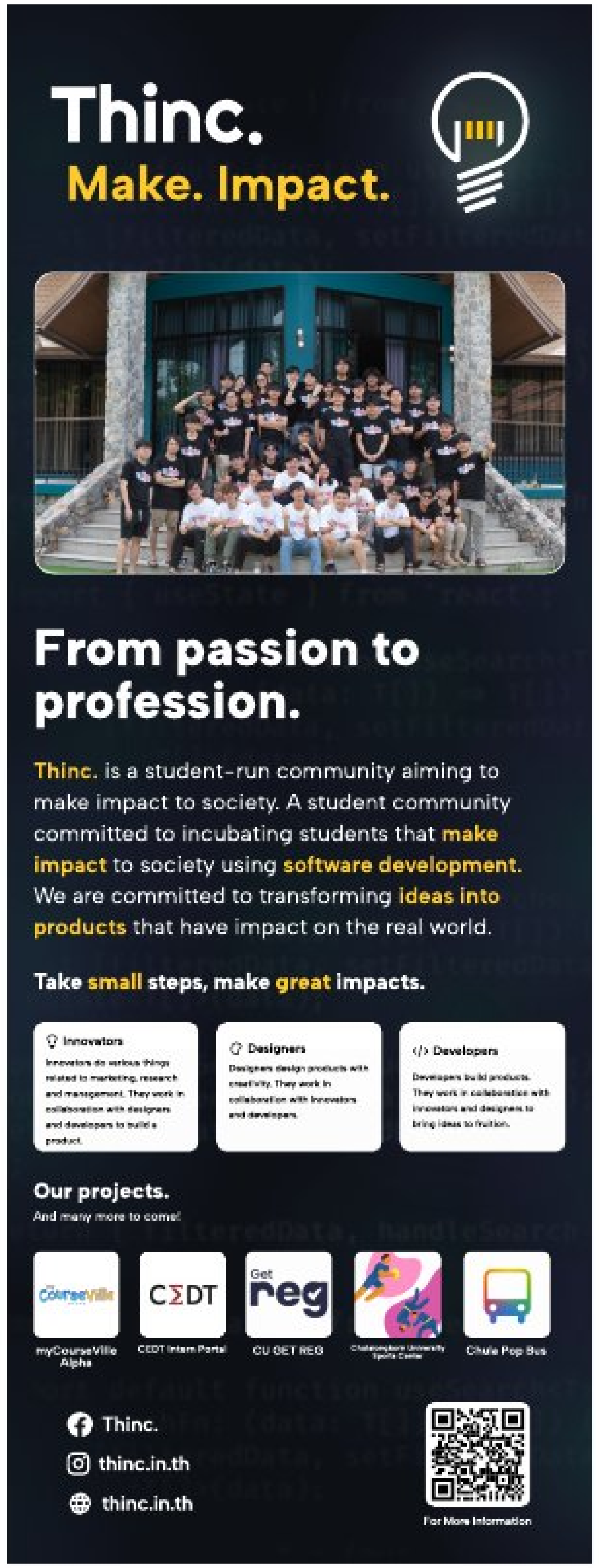
Thinc. Club

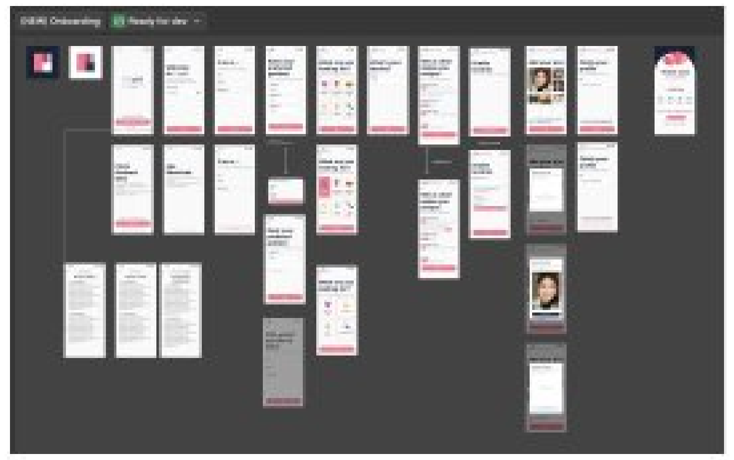
Role: Graphic Designer

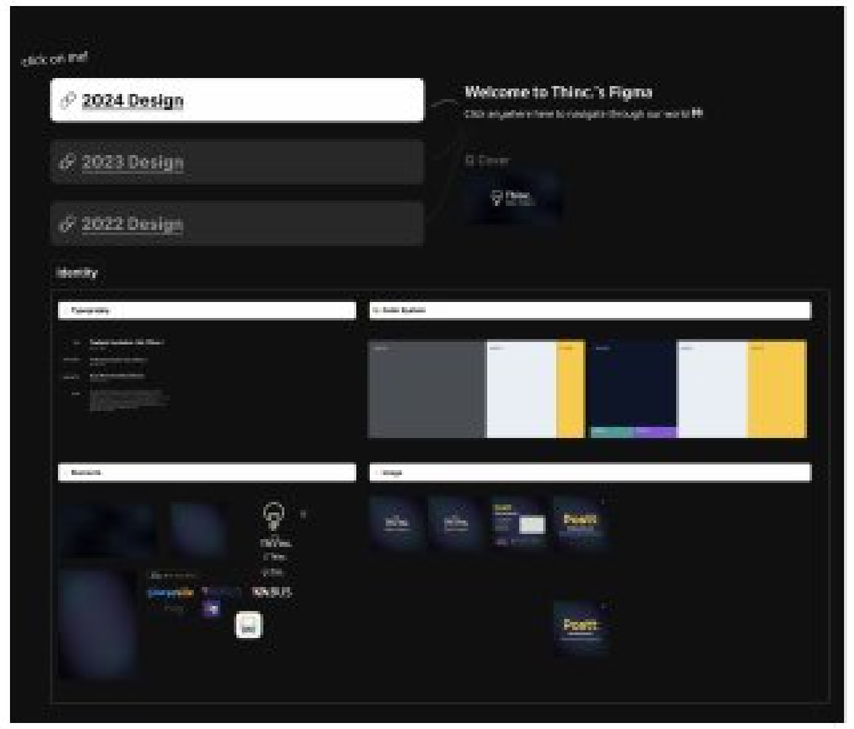
Thailand Incubator Club is a student-run community aiming to make impact to the real world.

I re-branded the club by using new typefaces and color schemes that reflect on the innovative nature of the club.

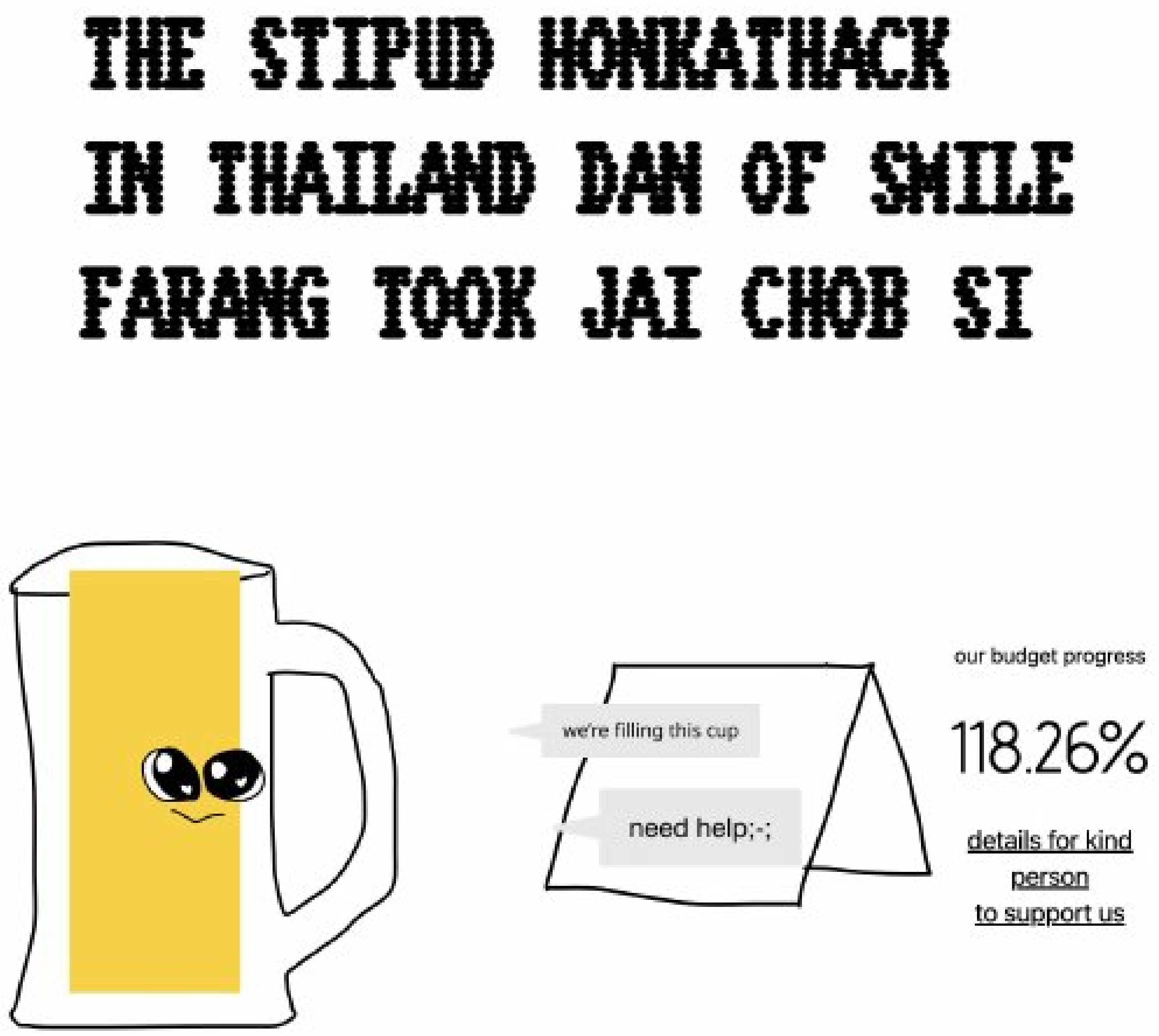
As a club member, I was the main designer of the poster from club events and was involved in the UX/UI process of several projects, such as CUpid, a Chula-exclusive dating app.







Miscellaneous Projects



7th and 8th Stupid Hackathon in Thailand

Role: Event Lead

I was the event director of the 7th and 8th Stupid Hackathon in Thailand, an event aimed at developers and designers wanting a Hackathon that enables real projects to be built in a humorous manner.

Managing a 3-day hackathon in 2 different venues is a challenging task which taught me on how to handle spontaneous stressful situations and enabled me to connect with tons of cool people that participated in the event.